

Explainable AML Transaction-Risk Triage for SME and Corporate Banking

Final report — Pillar 5 Capstone Project

Author: Julius Pabular · Date: 2026-09-08 · Released model: 20260904T225142-0dc8f82-hgb · Repository: <https://github.com/joopabs/aml-risk-triage-capstone>

Educational decision-support prototype trained on synthetic PaySim data. Outputs are risk scores and review priorities that help human investigators decide what to review first. This system makes no fraud or AML determination and performs no automatic blocking, account closure, customer risk rating, or regulatory reporting. Results on synthetic data do not establish real-world detection effectiveness, fairness, or regulatory suitability.

PaySim is synthetic mobile-money transaction data. It is not real SME, corporate, or Philippine banking data.

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1. Problem Statement

Business context

Banks serving small-and-medium enterprise (SME) and corporate clients monitor large volumes of payment transactions for possible money-laundering or fraud indicators. Investigator capacity is fixed and far smaller than transaction volume, so most alerts are never reviewed in depth and many reviewed alerts are false positives. This capstone builds an **educational decision-support prototype** that assigns each transaction a risk score and ranks transactions so that a limited number of daily investigator reviews is spent on the transactions most worth a human look.

The prototype is trained and evaluated on **PaySim**, a public synthetic mobile-money dataset with a simulated fraud label (Kaggle ealaxi/paysim1, CC BY-SA 4.0). PaySim stands in because real SME/corporate transaction data with labels is not publicly available. Every result is therefore about simulated fraud in synthetic data and is described as such throughout.

Intended decision, use, and non-use

Decision supported: which transactions, out of all transactions in a review period, should an investigator look at first, given a fixed capacity of K reviews? The system outputs a ranked list. A human decides what, if anything, to do with each reviewed transaction.

Human review workflow (FR-083). Each simulated day the system produces a queue of the K highest ranked transactions with a risk score, a review priority (high / medium / low), the three factors that moved the score most, the model version, and the disclaimer. An investigator works the queue, records a decision for each item, and may override the ranking or pull any transaction into review. Nothing happens to a transaction or an account unless a person acts. Override and decision records feed the monitoring plan and, in any real deployment, the fairness audit.

Explicit non-use. The system does not and must not: automatically block, hold or reverse transactions; close, freeze or restrict accounts; assign a customer- or entity-level risk rating; generate or file suspicious-activity or regulatory reports; make an actual AML or fraud determination; or be used on real customer data without a governance-controlled validation and fairness audit. The positive label is "simulated fraud"; model outputs are "risk scores" and "review priorities".

Task definition and success metrics

Item	Definition
Unit of analysis	One synthetic financial transaction (one PaySim row)
Task type	Binary classification producing a probability-like risk score used for ranking
Target	isFraud (1 = simulated fraud, 0 = simulated normal)
Prediction time	End-of-period batch triage: the transaction's own fields and posted balances are available; aggregates use only strictly earlier transactions
Primary technical metric	PR-AUC on the held-out temporal test split
Operational metric	Recall@K per review period, K = 200 (about 1% of median daily volume in the validation and test periods, and below the median of 265 positives per test day, so capacity binds)
Review period	24 steps (one simulated day)
Secondary metrics	Precision@K, precision, recall, F1, ROC-AUC, confusion matrix at the operating point, PR and calibration curves, Brier, ECE, false-positive rate
Business KPI (illustrative)	Positives surfaced within K daily reviews, and the improvement factor versus random selection and a simple rule ranking. Never expressed in currency.

Results against the success criteria (single-touch test evaluation, model 20260904T225142-0dc8f82-hgb)

- SC-001: Recall@200 = **0.7568** (mean over review periods; pooled 95% bootstrap CI [0.7252, 0.7866]) exceeds random ranking (0.1012) and the chronological dummy (0.2076); PR-AUC = **1.0000** exceeds the no-skill value of 0.0109.

- SC-002: the rule comparator (flag, then amount) reaches $\text{Recall}@200 = 0.3101$; the selected model exceeds it.
- Illustrative KPI: 200 positives surfaced per simulated day at $K = 200$, versus 82 for the rule ranking and 27 for random selection (illustrative counts on synthetic data).

$\text{Recall}@K$ is a ceiling set by capacity: every test period holds more than K positives and the top K are all positives. The near-perfect separability is a property of the PaySim generator (Sections 4–7), not evidence of real-world AML capability.

2. Dataset Overview and Data Dictionary

Data Quality Report

Scope

Source: `data/raw/PS_20174392719_1491204439457_log.csv`. Rows: 6,362,620. Columns: 11. All figures below are aggregates; no row-level data is shown.

Columns and nulls

column	dtype	nulls
step	int32	0
type	category	0
amount	float32	0
nameOrig	string	0
oldbalanceOrg	float32	0
newbalanceOrig	float32	0
nameDest	string	0
oldbalanceDest	float32	0
newbalanceDest	float32	0
isFraud	int8	0
isFlaggedFraud	int8	0

Duplicates

kind	count
exact duplicate rows	0
near-duplicates ignoring identifiers	543

Numeric summary and outliers (IQR rule)

column	min	p50	p95	p99	p99.9	max	zeros	negatives	IQR outliers
amount	0.0000	74,871.94	518,634.19	1,615,979.50	8,956,797.63	92,445,520.00	16	0	338,078
oldbalanceOrg	0.0000	14,208.00	5,823,702.10	16,027,256.35	26,825,930.48	59,585,040.00	2,102,449	0	1,112,507
newbalanceOrig	0.0000	0.0000	5,980,262.37	16,176,160.39	26,971,659.15	49,585,040.00	3,609,566	0	1,053,391
oldbalanceDest	0.0000	132,705.66	5,147,229.70	12,371,819.49	34,392,787.85	356,015,904.00	2,704,388	0	786,135
newbalanceDest	0.0000	214,661.45	5,515,715.97	13,137,866.93	39,531,570.22	356,179,264.00	2,439,433	0	738,527

Amount quantiles by transaction type

type	p50	p95	p99	p99.9	max
CASH_IN	143,427.71	412,005.12	550,870.85	727,791.33	1,915,267.88
CASH_OUT	147,072.19	427,877.16	579,654.09	864,676.47	10,000,000.00
DEBIT	3,048.99	14,795.35	50,817.98	179,208.45	569,077.50
PAYMENT	9,482.19	37,835.66	59,500.11	89,701.44	238,637.98
TRANSFER	486,308.38	2,674,586.95	10,000,000.00	24,840,512.69	92,445,520.00

Invalid values

check	count
zero amount	16
negative amount	0
negative oldbalanceOrig	0
negative newbalanceOrig	0
negative oldbalanceDest	0
negative newbalanceDest	0
origin balance arithmetic inconsistent (tol 0.01)	3,953,846
destination balance arithmetic inconsistent (tol 0.01)	5,229,368

Balance arithmetic by type

type	n	orig inconsistent	rate	dest inconsistent	rate	orig zero after	dest both zero
CASH_IN	1,399,284	234,237	0.1674	1,399,284	1.0000	0	160,005
CASH_OUT	2,237,500	2,007,949	0.8974	1,324,750	0.5921	959,412	1,608
DEBIT	41,432	12,660	0.3056	23,467	0.5664	5,628	0
PAYMENT	2,151,495	1,187,615	0.5520	2,151,495	1.0000	326,879	2,151,495
TRANSFER	532,909	511,385	0.9596	330,372	0.6199	228,662	4,174

Class imbalance

metric	value
positives (simulated fraud)	8,213
negatives	6,354,407
prevalence	0.0013
negatives per positive	773.7011

type	n	positives	rate
CASH_IN	1,399,284	0	0.0000
CASH_OUT	2,237,500	4,116	0.0018
DEBIT	41,432	0	0.0000
PAYMENT	2,151,495	0	0.0000
TRANSFER	532,909	4,097	0.0077

Time steps

metric	value
steps observed	743
step range	1–743
transactions per step (min / median / max)	2 / 529 / 51,352
positives per step (min / median / max)	0 / 10 / 40
steps with zero positives	2
first / last step with a positive	1 / 743

Per-step counts are in `data_quality.json` under `steps.transactions_by_step` and `steps.cumulative_positives_by_step` (used to choose split bounds, V9).

Identifiers

column	unique	share appearing >1	max occurrences
nameOrig	6,353,307	0.0015	3
nameDest	2,722,362	0.1688	113

Rule flag column

column	flagged	rate	flagged and positive	precision as rule	recall as rule
isFlaggedFraud	16	0.0000	16	1.0000	0.0019

Sensitive-attribute pre-scan

Column names checked against: age, gender, sex, ethnicity, race, nationality, income, socioeconomic, region, zip, postcode, birth, religion, marital.

No column name matches a sensitive-attribute pattern. The formal availability record is produced in Milestone 7 (FR-070).

Findings and handling decisions

Written after reviewing the generated tables (task T022, 2026-09-05). Every number below is copied from reports/data_quality.json. Decisions are one of keep, correct, flag as feature, exclude.

ID	Finding (from the tables above)	Decision	Justification
DQ-01	6,362,620 rows, 11 columns, 0 nulls in every column.	keep	Nothing to impute; the imputation step in the pipeline is a no-op guard only.
DQ-02	0 exact duplicate rows; 543 near-duplicate pairs when identifiers are ignored (1,081 rows, 27 of them positives).	keep	Identical amounts and balances with different accounts are distinct transactions in a simulator with coarse value grids; dropping them would delete real positives.
DQ-03	16 zero-amount rows, all CASH_OUT, and all 16 are positives.	flag as feature, treat as artifact	A zero-amount transaction has no economic meaning; a perfect label correlation on 16 rows is a simulator artifact. zero_amount_flag is kept as a feature but the report must state that it contributes leakage-like lift, and the strict feature set is evaluated with and without it (research R-06).
DQ-04	No negative amounts or balances.	keep	Hard minimums hold; soft minimums record zero violations.
DQ-05	Origin balance arithmetic (direction-aware, tolerance 0.01) is inconsistent for 89.7% of CASH_OUT, 96.0% of TRANSFER, 55.2% of PAYMENT, 30.6% of DEBIT and 16.7% of CASH_IN rows.	flag as feature	Inconsistency is simulator behavior, not an error to correct. orig_balance_inconsistent_flag and the signed gap become batch-only features; their importance is reported as artifact evidence.
DQ-06	Destination balance arithmetic is inconsistent for 100% of PAYMENT and CASH_IN rows and 59–62% of CASH_OUT/TRANSFER rows. All 2,151,495 PAYMENT destinations are merchants (identifier prefix M), all with both destination balances equal to zero, and none of them is a positive.	flag as feature	Merchants carry no balance state in PaySim. dest_is_merchant (derived from the identifier prefix, not the identifier itself) and dest_balance_inconsistent_flag are batch-only features. FR-033 is respected because account identity is never used.
DQ-07	Heavy tails: amount p50 74,872 vs max 92,445,520; 338,078 IQR outliers; TRANSFER p99 is exactly 10,000,000. 5,650 rows at or above 10,000,000 contain 287 positives and only 3 rule flags.	keep, transform	Positives live in the tails, so no removal. Use log1p(amount) and training-fitted amount buckets; report per-type quantiles.

ID	Finding (from the tables above)	Decision	Justification
DQ-08	8,213 positives, prevalence 0.129%, 774 negatives per positive.	keep, handle in training	Class weighting or in-fold undersampling only (FR-036); PR-AUC primary; accuracy never a headline (FR-007).
DQ-09	Positives occur only in CASH_OUT (4,116) and TRANSFER (4,097). CASH_IN, DEBIT and PAYMENT have zero positives across 3.59 million rows.	keep all types	The model must score every type; type is a first-order feature. Ablations on TRANSFER/CASH_OUT only are reported as a sensitivity check, not as the primary result.
DQ-10	Steps 1–743 are contiguous. Transaction volume is highly non-stationary: 574,255 transactions in day 1 and 455,238 in day 2, then 1,070 in day 3; days 6–17 carry roughly 350,000–450,000 each; days 18–31 carry 272 to 57,853. Positives are almost constant at 216–320 per day regardless of volume; day 31 (steps 721–743) has 272 transactions, all positive.	keep, constrain split design	Simulated fraud is injected at a near-constant rate independent of volume, so prevalence in late steps is orders of magnitude above early steps. The temporal split (T025) must be chosen so validation and test periods contain enough transactions per review period for Recall@K to mean anything, and the report must present validation-vs-test shift explicitly.
DQ-11	Origin identifiers are almost never reused (6,353,307 unique; 0.15% appear more than once; max 3). Destination identifiers are reused (2,722,362 unique; 16.9% appear more than once; max 113).	keep, guide features	Prior-transaction aggregates per origin are expected to be uninformative; destination aggregates are the causal-aggregate candidates (V10 decides). Identifiers themselves are dropped after aggregation.
DQ-12	Rule flag <code>isFlaggedFraud</code> fires 16 times, all on positives: precision 1.000, recall 0.0019.	keep as comparator only	Too sparse to rank a queue on its own; the rule comparator ranks flagged rows first then by amount descending (research R-10). Never a model feature.
DQ-13	No column name matches a sensitive-attribute pattern.	record	The formal availability record (FR-070) is produced in Milestone 7; the expected path is the operational error-slice analysis.

Source-data limitations

- **Synthetic origin.** PaySim is generated by an agent-based simulator seeded with aggregate statistics from a mobile-money service. Nothing here is real SME, corporate, or Philippine banking data, and no result can establish real-world detection effectiveness, fairness, or regulatory suitability.
- **Constant fraud injection (DQ-10).** Positives arrive at a near-constant daily rate while legitimate volume swings by three orders of magnitude. Late-period prevalence is therefore artificially high, and any temporal test split inherits that shift. This is the single largest threat to interpreting Recall@K as an operational figure.
- **Label-correlated artifacts (DQ-03, DQ-05, DQ-06).** Zero amounts, balance-arithmetic gaps and zero-balance destinations correlate with the label because of how the simulator writes state, not because of transferable behavior. Features built from them are kept for the batch-triage framing but must be reported as artifact-driven lift, with a strict feature set as the comparison.
- **Merchant state is absent (DQ-06).** Merchant destinations have no balances, so destination features are uninformative for PAYMENT and only meaningful for customer-to-customer types.
- **Two positive types only (DQ-09).** Results say nothing about detecting simulated fraud in CASH_IN, DEBIT or PAYMENT, because none exists in the data.
- **Coarse time index.** `step` is an hourly index with no calendar, weekday or intraday semantics beyond position; period boundaries for Recall@K are an analytical convention (24 steps), not an observed business day.

- **No demographics (DQ-13).** Demographic fairness cannot be measured; only operational error slices can be reported, and they must not be described as fairness across protected groups.

Split summary (validation task V9)

Strategy: **temporal** by step; config hash sha256:cac2e0e01afa...; review period 24 steps; no rows excluded (every DQ finding was keep or flag). Figures copied from data/processed/split_manifest.json.

split	steps	days	rows	positives	prevalence
train	1–408	1–17	5,987,417	4,589	0.0766%
val	409–552	18–23	181,068	1,504	0.8306%
test	553–743	24–31	194,135	2,120	1.0920%

Why these bounds: validation and test both sit in the low-volume regime (DQ-10), so the operating point chosen on validation applies to a test period with the same volume and prevalence character. Training spans both regimes (days 3–5 are also low-volume). Each split keeps at least 1,500 positives, three times the min_positives_per_split guard of 500. The prevalence jump from 0.08% in training to about 1% in validation and test is a property of the simulator, not of the split, and the report must present validation-versus-test shift explicitly (FR-045).

Review capacity $K = 200$ is about 1% of the median daily volume across the validation and test days (20,363) and below the median positives per day (264), so capacity binds and Recall@K is a real constraint rather than a formality. The k_grid [50, 100, 200, 300, 500] shows sensitivity.

Data Dictionary

Conventions

availability: realtime = known when the transaction is observed; batch_only = known in end-of-period batch triage (post-transaction state, research R-06); label = target, never an input. Identifiers are never model features (FR-033). Observed ranges come from reports/data_quality.json; [PROFILE] means profiling has not run yet.

Raw variables

variable	type	unit	range / allowed values	role	prediction-time availability	description
step	int32	hours since simulation start (expected; V8 confirms)	1 – 743	time_index	realtime	Simulation time step; one unit is expected to represent one hour.
type	category	category	CASH_IN, CASH_OUT, DEBIT, PAYMENT, TRANSFER	feature	realtime	Transaction type (expected values include CASH_IN, CASH_OUT, DEBIT, PAYMENT, TRANSFER; V2 confirms).
amount	float32	currency units (simulated)	0.00 – 92,445,520.00	feature	realtime	Transaction amount in the simulator's currency units.
nameOrig	string	identifier	6,353,307 unique	identifier	realtime	Origin account identifier. Never a model feature; used only for causal aggregates.
oldbalanceOrg	float32	currency units (simulated)	0.00 – 59,585,040.00	feature	realtime	Origin account balance before the transaction.

variable	type	unit	range / allowed values	role	prediction-time availability	description
newbalanceOrig	float32	currency units (simulated)	0.00 – 49,585,040.00	feature	batch_only	Origin account balance after the transaction (post-transaction state).
nameDest	string	identifier	2,722,362 unique	identifier	realtime	Destination account identifier. Never a model feature; used only for causal aggregates.
oldbalanceDest	float32	currency units (simulated)	0.00 – 356,015,904.00	feature	realtime	Destination account balance before the transaction.
newbalanceDest	float32	currency units (simulated)	0.00 – 356,179,264.00	feature	batch_only	Destination account balance after the transaction (post-transaction state).
isFraud	int8	binary flag	0, 1	target	label	Simulated fraud label; 1 = simulated fraud, 0 = simulated normal transaction.
isFlaggedFraud	int8	binary flag	0, 1	rule_comparator	batch_only	Simulator rule flag. Never a model feature; defines the rule comparator ranking.

Engineered features

variable	type	unit	range / allowed values	role	prediction-time availability	description
type_onehot	one-hot (5 columns)	indicator	0/1 per observed type	categorical	realtime	One column per transaction type observed in training; unknown types map to all zeros. Rationale: Simulated fraud occurs only in TRANSFER and CASH_OUT (DQ-09); type is the first-order signal.
log_amount	float	log(currency units + 1)	>= 0	numeric	realtime	log1p(amount). Rationale: Amounts are heavy-tailed (DQ-07); log scale stabilises linear models and keeps tree splits interpretable.
amount_bucket	int	bucket index	0..n_bins-1 (n_bins=10)	numeric	realtime	Amount decile bucket; edges fitted on the training split only. Rationale: Quantile buckets fitted on training data give a monotone, outlier-robust view of amount for slice analysis and linear models.

variable	type	unit	range / allowed values	role	prediction-time availability	description
log_oldbalance_org	float	log(currency units + 1)	>= 0	numeric	realtime	log1p(oldbalanceOrg). Rationale: Origin balance before the transaction is heavy-tailed with a mass at zero (DQ-07); log scale plus a zero flag captures both.
log_oldbalance_dest	float	log(currency units + 1)	>= 0	numeric	realtime	log1p(oldbalanceDest). Rationale: Destination balance before the transaction; merchants carry zero (DQ-06), customers vary widely.
amount_to_orig_balance_ratio	float	ratio	>= 0	numeric	realtime	amount / (oldbalanceOrg + 1). Rationale: Emptying an account (ratio near 1) is a classic mule-account pattern; the +1 guard handles zero balances.
orig_zero_balance_flag	int	indicator	0/1	flag	realtime	oldbalanceOrg == 0. Rationale: 2.1M rows start from a zero origin balance (DQ-07); the flag separates this mass from the continuous scale.
dest_zero_balance_flag	int	indicator	0/1	flag	realtime	oldbalanceDest == 0. Rationale: Zero destination balance before the transaction marks merchants and fresh accounts (DQ-06).
zero_amount_flag	int	indicator	0/1	flag	realtime	amount == 0. Simulator artifact (DQ-03). Rationale: All 16 zero-amount rows are positives (DQ-03); kept as a feature but reported as a simulator artifact, never as transferable signal.
dest_is_merchant	int	indicator	0/1	flag	realtime	nameDest starts with 'M'. Derived account type; the identifier itself is dropped (FR-033). Rationale: Merchant destinations (identifier prefix M) receive only PAYMENT and carry no positives or balance state (DQ-06); this is account type, not identity.

variable	type	unit	range / allowed values	role	prediction-time availability	description
step_hour_of_day	int	hour	0..23	numeric	realtime	(step - 1) mod 24. Rationale: Hourly position within the simulated day may carry volume seasonality; cyclic and in-range for every split.
step_day_index	int	day	0..30	numeric	realtime	(step - 1) // 24. Excluded from every modeling set. Rationale: Diagnostic only. Under a temporal split the test days are out of the training range, so using it would encode regime rather than behaviour (DQ-10).
orig_balance_delta	float	currency units	any	numeric	batch_only	oldbalanceOrig - newbalanceOrig. Rationale: Posted change in origin balance; in batch triage the posted state is available and its mismatch with amount is informative (DQ-05).
dest_balance_delta	float	currency units	any	numeric	batch_only	newbalanceDest - oldbalanceDest. Rationale: Posted change in destination balance (DQ-06).
orig_balance_inconsistent	int_flag	indicator	0/1	flag	batch_only	newbalanceOrig - expected > 0.01 where expected = old + amount for CASH_IN, old - amount otherwise. Rationale: Direction-aware arithmetic gap on the origin side is inconsistent for most CASH_OUT/TRANSFER rows (DQ-05); a simulator behaviour that correlates with the label.
dest_balance_inconsistent	int_flag	indicator	0/1	flag	batch_only	newbalanceDest - oldbalanceDest - amount > 0.01. Rationale: Destination arithmetic gap (DQ-06); always inconsistent for merchant destinations.

variable	type	unit	range / allowed values	role	prediction-time availability	description
orig_zero_after_flag	int	indicator	0/1	flag	batch_only	oldbalanceOrig > 0 and newbalanceOrig == 0. Rationale: Account emptied to exactly zero after the transaction (DQ-05); a strong mule pattern in the simulator.
orig_prior_txn_count	int	count	>= 0	aggregate	realtime	Count of earlier transactions with the same nameOrig (strictly earlier step, or earlier file position within the same step). Rationale: Number of strictly earlier transactions by the same origin; expected near zero because origins rarely repeat (DQ-11), kept to let V10 confirm.
orig_prior_amount_sum	float	currency units	>= 0	aggregate	realtime	Sum of amount over earlier transactions with the same nameOrig. Rationale: Cumulative earlier outflow by the same origin (DQ-11).
dest_prior_txn_count	int	count	>= 0	aggregate	realtime	Count of earlier transactions with the same nameDest. Rationale: Destinations repeat (16.9% appear more than once, max 113; DQ-11); accumulation into one destination is a laundering-style pattern.
dest_prior_amount_sum	float	currency units	>= 0	aggregate	realtime	Sum of amount over earlier transactions with the same nameDest. Rationale: Cumulative earlier inflow to the same destination (DQ-11).

3. EDA and Feature Engineering Report

EDA Summary

Scope

Figures below use the **training split** (steps 1–408) unless labeled descriptive; class-conditional plots use all training positives plus a seeded sample of negatives. No modeling decision here uses validation or test rows. Observations are written by a human in `reports/eda_narrative.md` after viewing each figure (task T036).

Training split: rows and positives by type

type	normal	positive	positive rate
CASH_IN	1,313,389	0	0.0000
CASH_OUT	2,115,988	2,304	0.0011
DEBIT	38,489	0	0.0000
PAYMENT	2,019,717	0	0.0000
TRANSFER	495,245	2,285	0.0046

Flag features: positive rate by value (training split)

flag	value	n	positive rate
orig_zero_balance_flag	0	3,990,733	0.0011
orig_zero_balance_flag	1	1,996,684	0.0000
dest_zero_balance_flag	0	3,452,391	0.0005
dest_zero_balance_flag	1	2,535,026	0.0012
zero_amount_flag	0	5,987,413	0.0008
zero_amount_flag	1	4	1.0000
dest_is_merchant	0	3,967,700	0.0012
dest_is_merchant	1	2,019,717	0.0000
orig_balance_inconsistent_flag	0	1,682,688	0.0027
orig_balance_inconsistent_flag	1	4,304,729	0.0000
dest_balance_inconsistent_flag	0	1,038,716	0.0011
dest_balance_inconsistent_flag	1	4,948,701	0.0007
orig_zero_after_flag	0	4,564,614	0.0000
orig_zero_after_flag	1	1,422,803	0.0032

Class balance by type

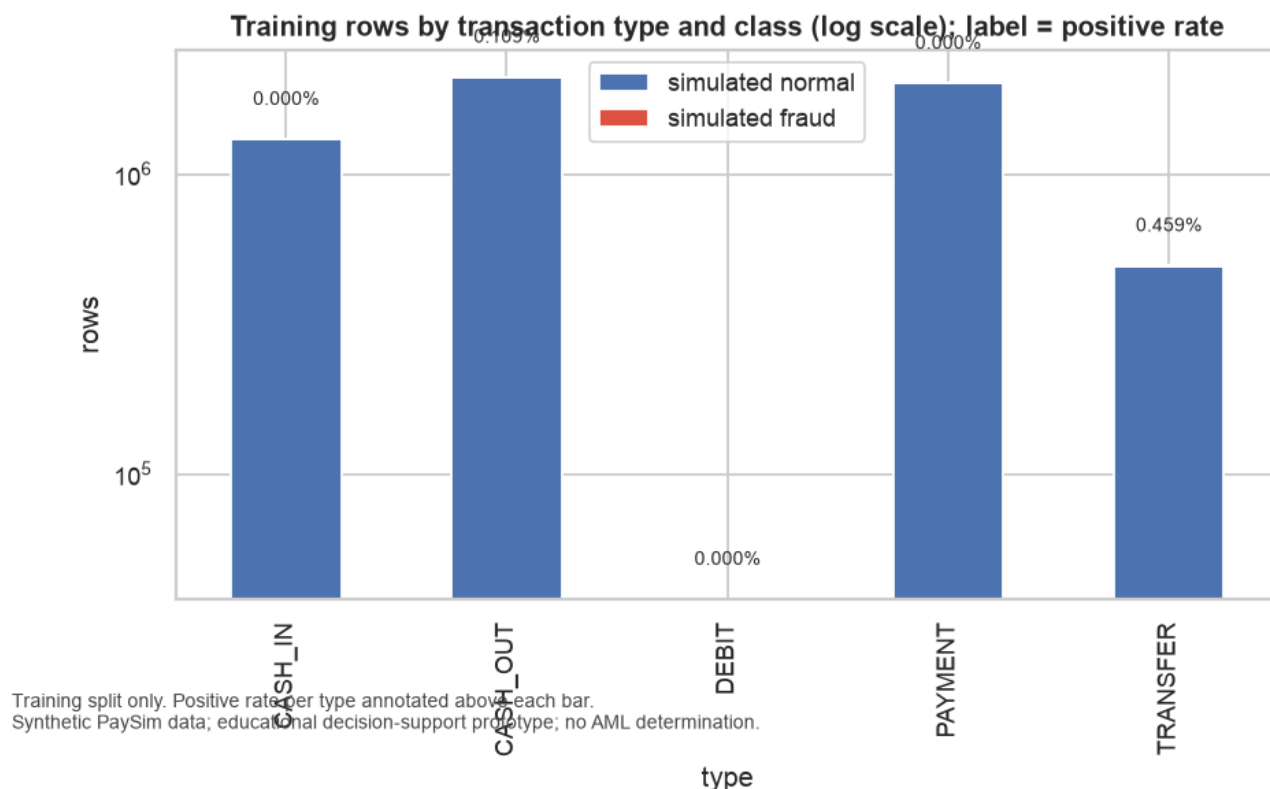


Figure: eda_01_class_by_type.png

Observation: Training positives exist only in CASH_OUT (2,304 of 2,118,292 rows, 0.109%) and TRANSFER (2,285 of 497,530, 0.459%). CASH_IN, DEBIT and PAYMENT carry zero positives across 3.37 million rows. Type is therefore the first-order feature, and TRANSFER has about four times the positive rate of CASH_OUT. Any model must still score the three positive-free types, which will dominate the low end of the ranking.

Amount by class

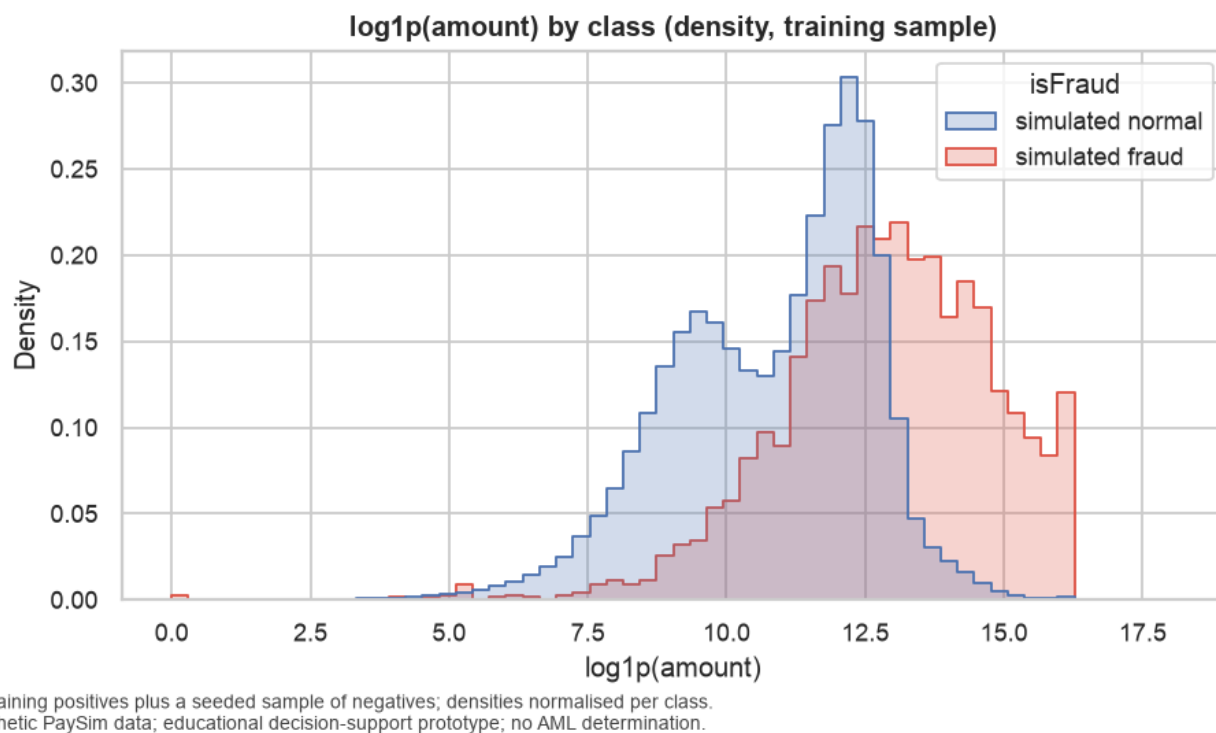
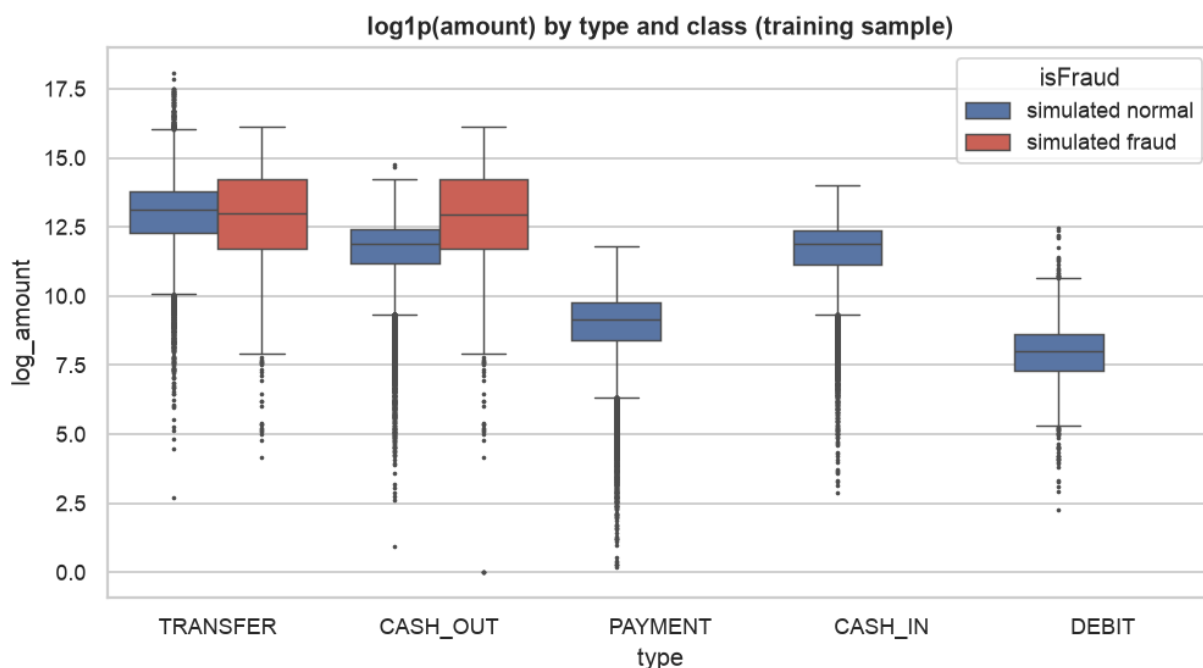


Figure: eda_02_amount_by_class.png

Observation: Positives sit to the right of normals on log1p(amount): their mode is around 13–14 versus a bimodal normal distribution peaking near 9.5 and 12.3. Two spikes are artifacts rather than behaviour: a narrow spike at

about 16.1, which is $\log_{10}$ of the 10,000,000 cap seen in the TRANSFER quantiles (DQ-07), and a small spike at 0 from the zero-amount rows (DQ-03). `log_amount` is informative but its extreme values are simulator boundaries.

Amount by type and class

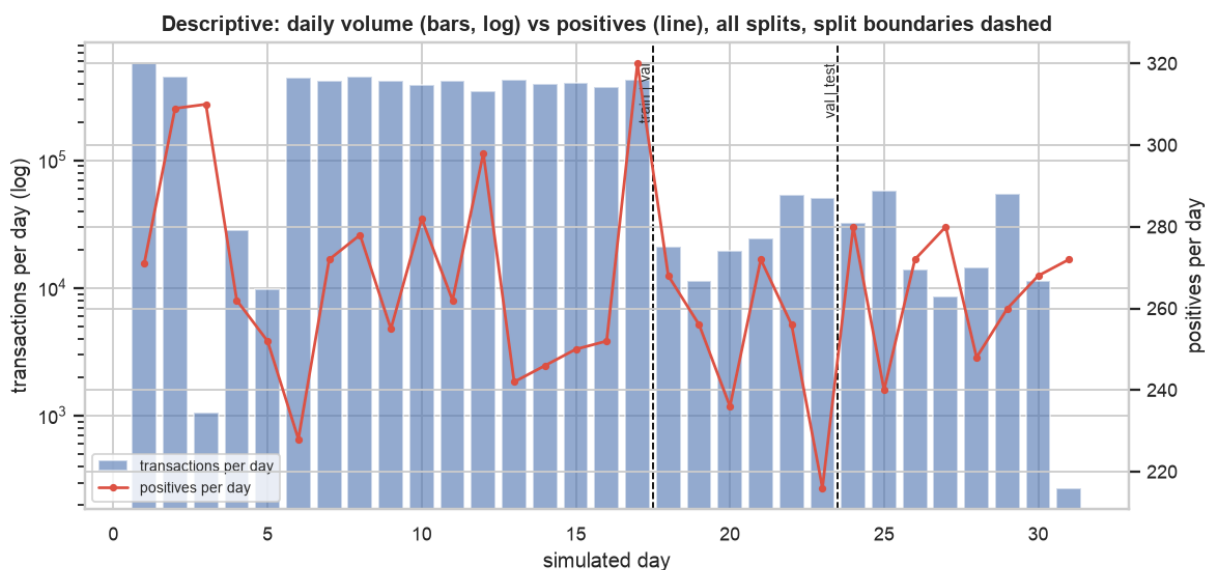


Box plots on the training sample; types without positives show a single box.
Synthetic PaySim data; educational decision-support prototype; no AML determination.

Figure: eda_03_amount_by_type_class.png

Observation: Within CASH_OUT the positive median amount (about 13 on the log scale) is a full log unit above the normal median (about 12). Within TRANSFER the medians are similar, near 13, but positives have a wider interquartile range and their maximum stops at the 16.1 cap while normal TRANSFER amounts reach about 18. Amount separates classes inside CASH_OUT much better than inside TRANSFER.

Volume and positives over time



Descriptive view across all splits (no modeling decision). Positives arrive at a near-constant rate while volume varies by three orders of magnitude (DQ-10).
Synthetic PaySim data; educational decision-support prototype; no AML determination.

Figure: eda_04_volume_positives_over_time.png

Observation: Daily volume runs at roughly 400,000–575,000 transactions on days 1–2 and 6–17 and collapses to about 1,000–58,000 on days 3–5 and 18–31, a change of nearly three orders of magnitude. Positives per day stay

between 216 and 320 throughout. This confirms DQ-10: simulated fraud is injected at a near-constant rate independent of volume. The split boundaries (dashed) place validation and test in the low-volume regime.

Prevalence by day

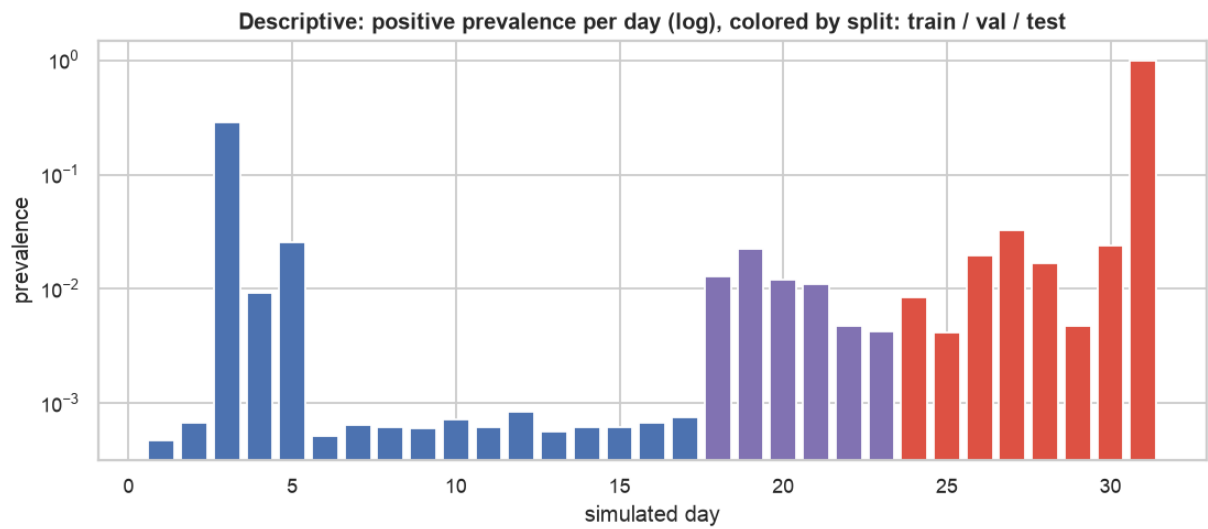


Figure: eda_05_prevalence_by_day.png

Observation: Prevalence on high-volume training days is 0.05%–0.07%. It jumps to about 29% on day 3 and 2.5% on day 5 (low-volume training days), sits between 0.4% and 2.3% across validation days, and between 0.4% and 3.3% across test days, with day 31 at 100% because only 272 transactions exist and all are positive. Validation and test prevalence are more than ten times training prevalence. Probability calibration fitted on training will be off in validation and test; this is why the operating point and any calibration are fitted on validation (FR-044, research R-09).

Correlation heatmap

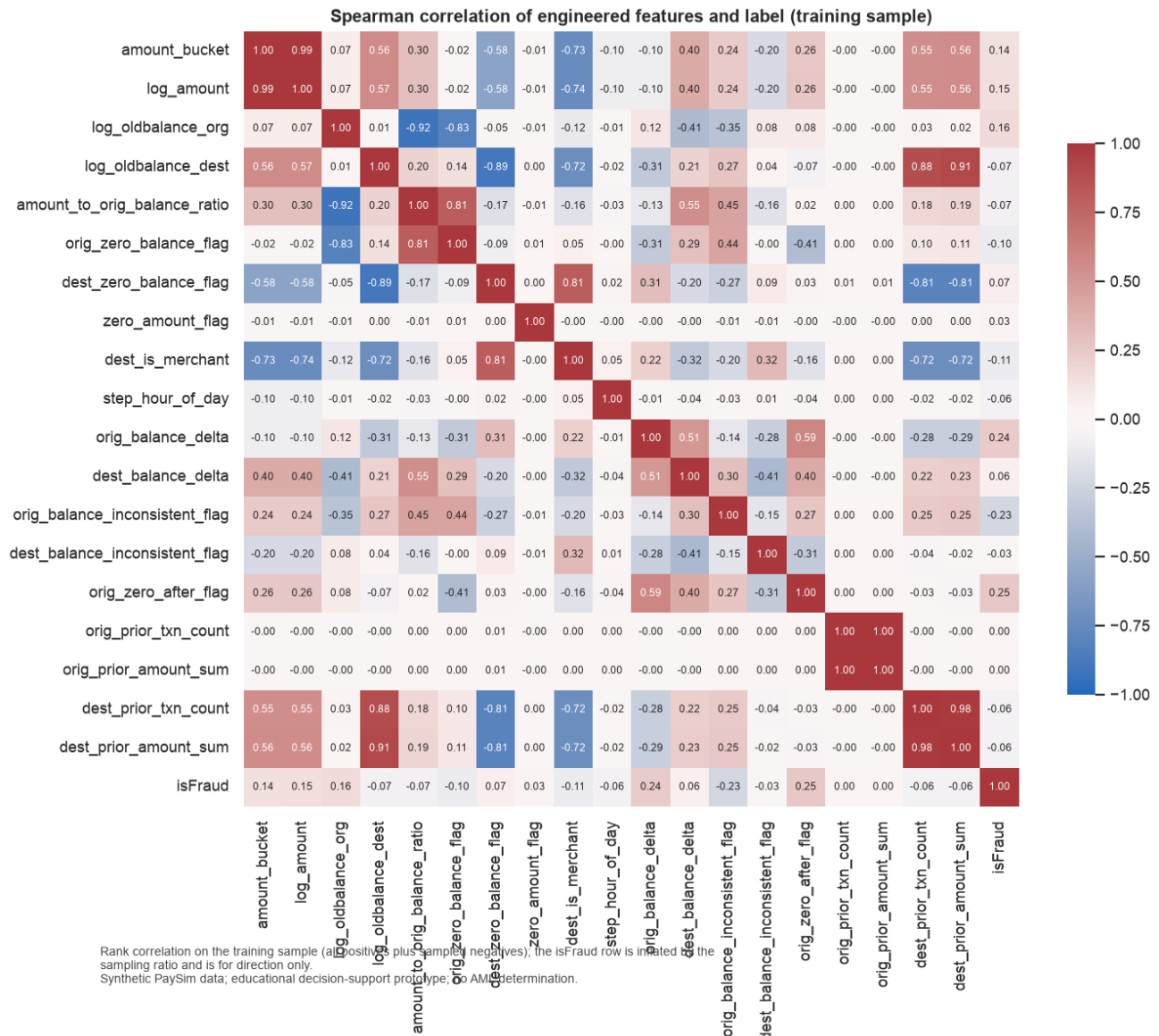


Figure: eda_06_correlation_heatmap.png

Observation: Rank correlations with the label (inflated by the positive-enriched sample, direction only): highest positive are orig_zero_after_flag (0.25), orig_balance_delta (0.24), log_oldbalance_org (0.16), log_amount (0.15); highest negative are orig_balance_inconsistent_flag (-0.23), dest_is_merchant (-0.11), orig_zero_balance_flag (-0.10). Origin aggregates correlate with nothing (0.00 everywhere), confirming DQ-11 that origins do not repeat. Strong collinearity to remember for feature selection: amount_bucket/log_amount (0.99), dest_prior_txn_count/dest_prior_amount_sum (0.98), log_oldbalance_dest with the destination aggregates (0.88–0.91) and with dest_zero_balance_flag (-0.89), amount_to_orig_balance_ratio with log_oldbalance_org (-0.92), and dest_is_merchant with dest_zero_balance_flag (0.81). PDP/ICE for these pairs will need caveats (FR-062).

Feature distributions by class

Class-conditional distributions of numeric features (training sample; blue normal, red positive)

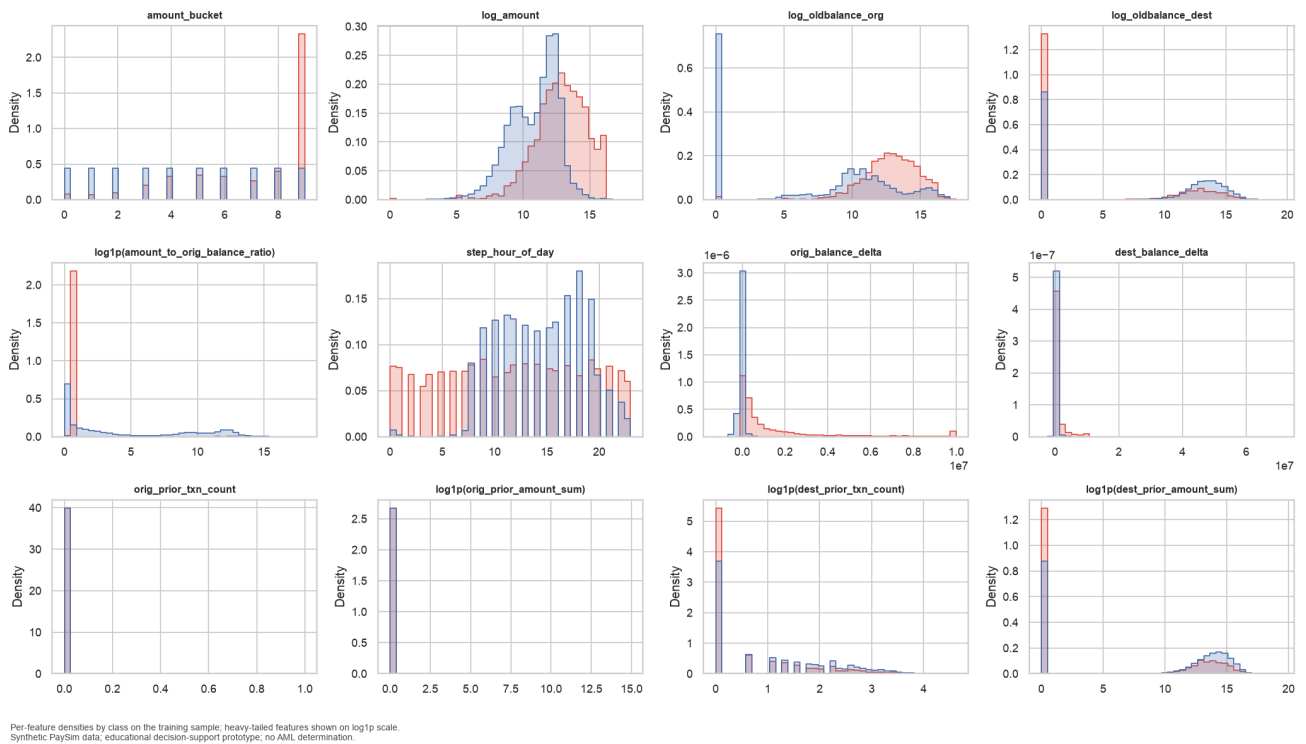


Figure: eda_07_feature_distributions.png

Observation: Positives concentrate in the top amount decile (bucket 9) at roughly five times the normal density. Normals have a large mass at zero origin balance while positives are centred near log 13, so positives come from funded accounts. `log1p(amount_to_orig_balance_ratio)` for positives spikes at about 0.69, which is a ratio of 1: the transaction empties the account. Positives are spread almost uniformly over the 24 hours while normals concentrate in hours 8–20. `orig_balance_delta` for positives has a long right tail to 10,000,000. `orig_prior_txn_count` and `orig_prior_amount_sum` are degenerate at zero for both classes and carry no information (candidates to drop in Milestone 4). Destination aggregates show positives more often going to a destination with no prior activity.

Flag positive rates

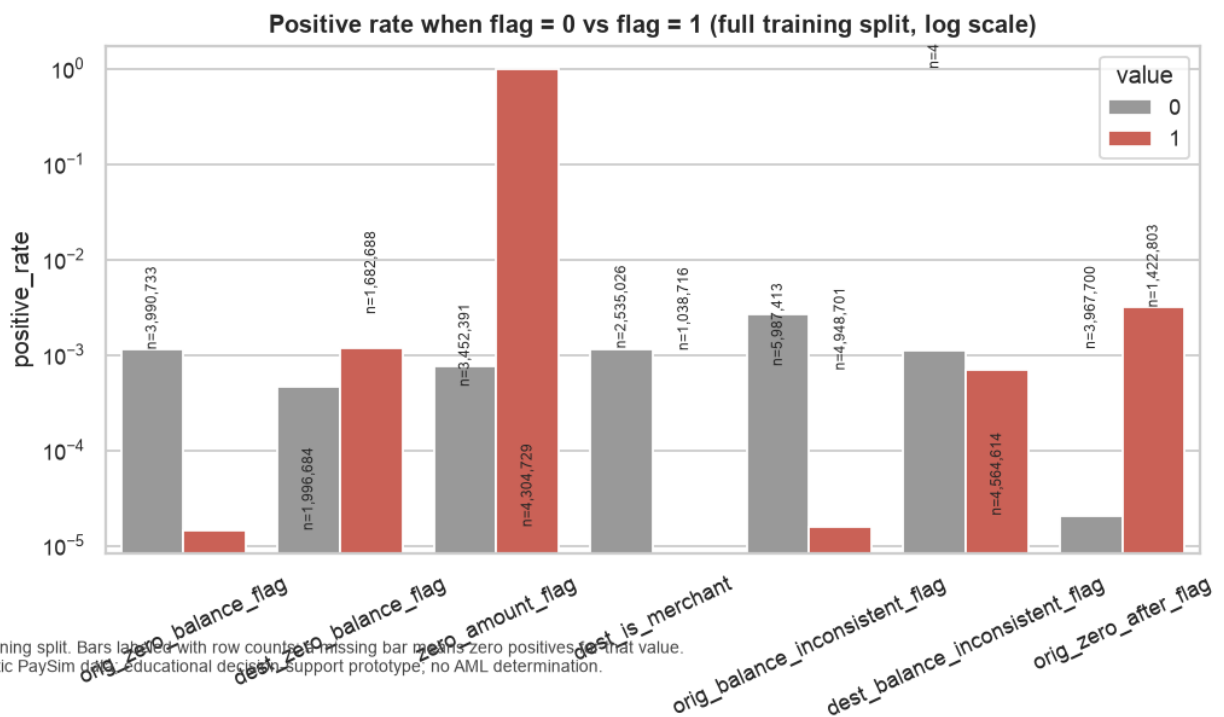
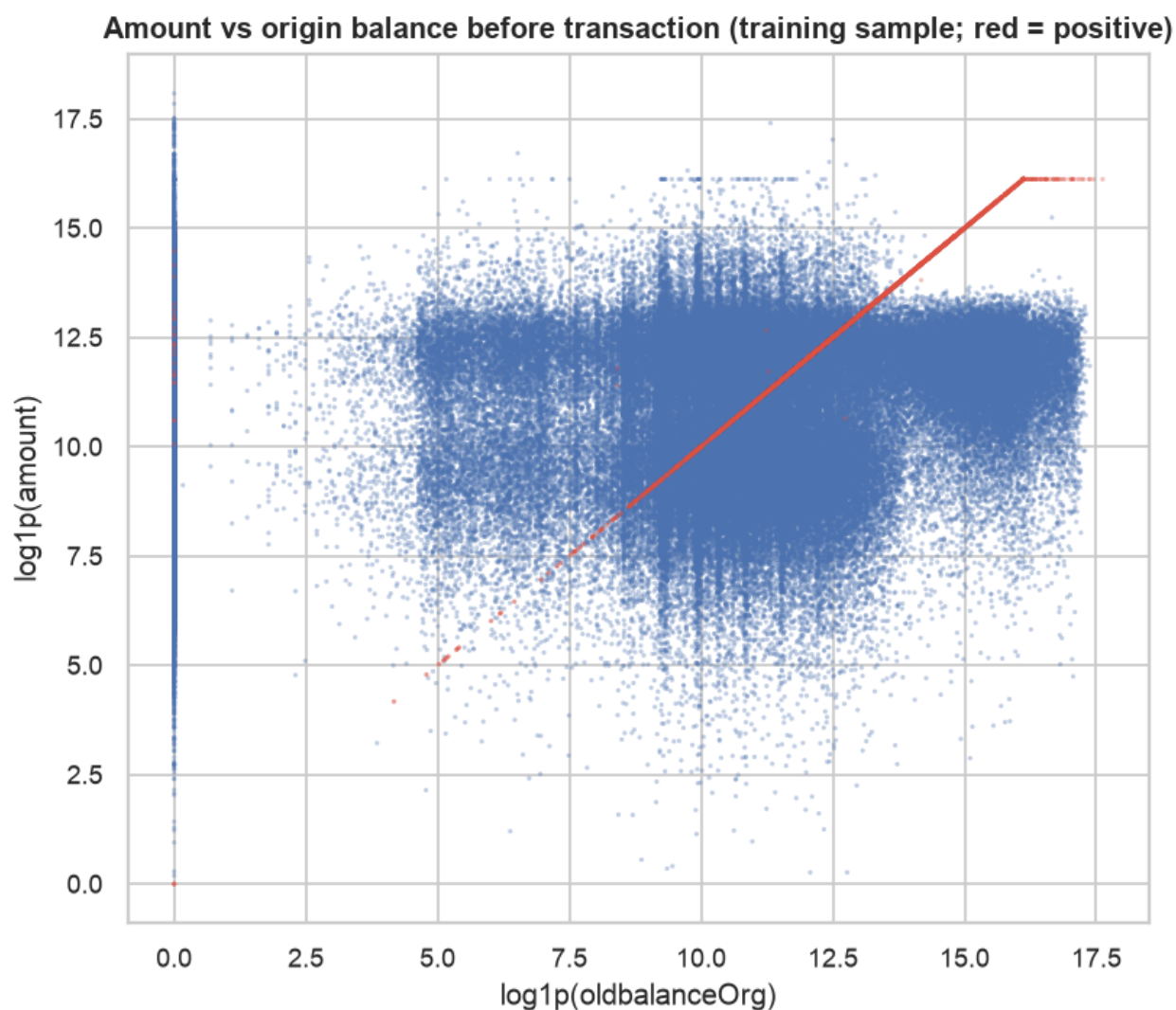


Figure: eda_08_flag_positive_rates.png

Observation: From the flag table: `zero_amount_flag = 1` has 4 training rows, all positive (DQ-03 artifact). `dest_is_merchant = 1` (2,019,717 rows) has zero positives. `orig_zero_balance_flag = 1` (1,996,684 rows) has a positive rate of 0.0015%, about 70 times lower than when the balance is non-zero. `orig_zero_after_flag = 1` has a rate of 0.32% versus 0.002% when 0, a factor of about 160. `orig_balance_inconsistent_flag = 1` (4,304,729 rows) has a rate of 0.0016% versus 0.27% when the arithmetic is consistent: simulated fraud rows have exact bookkeeping while most normal rows do not. This is the clearest simulator artifact among the batch-only features and the reason the strict pre-transaction set is evaluated alongside the primary set (research R-06). `dest_zero_balance_flag` and `dest_balance_inconsistent_flag` move the rate by less than a factor of three.

Amount vs origin balance

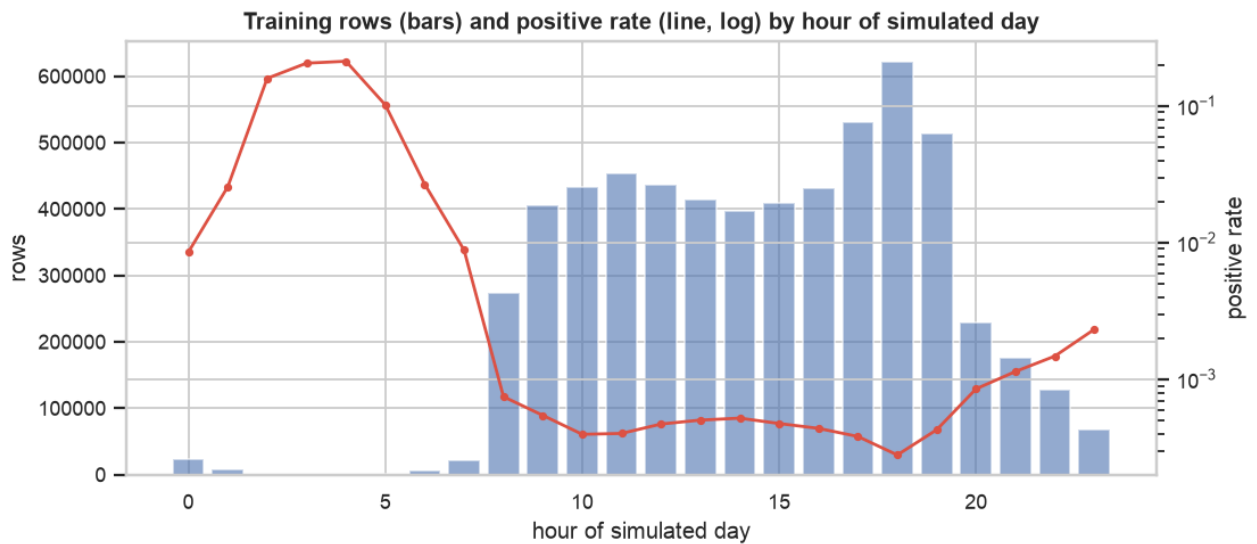


Training sample; positives drawn on top. The diagonal is `amount == oldbalanceOrg` (account emptied).
Synthetic PaySim data; educational decision-support prototype; no AML determination.

Figure: eda_09_amount_vs_origbalance.png

Observation: Positives lie almost exactly on the diagonal `amount = oldbalanceOrg` from about log 5 to log 16.1, then form a horizontal plateau at 16.1 where the 10,000,000 cap binds. Normals form a broad cloud plus a vertical band at zero balance. The "empty the account" pattern is the dominant visual signature of simulated fraud and is captured by `amount_to_orig_balance_ratio` and `orig_zero_after_flag`.

Hour of day

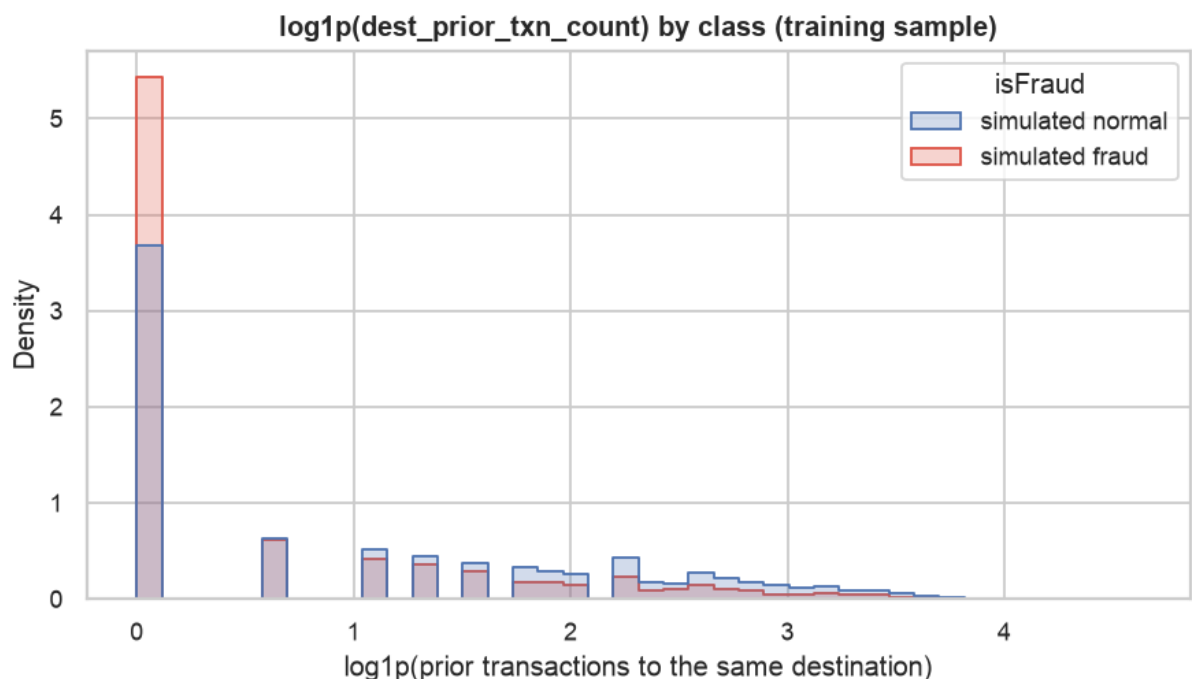


Training split only.
Synthetic PaySim data; educational decision-support prototype; no AML determination.

Figure: eda_10_hour_of_day.png

Observation: Hours 0–7 carry very few training rows (tens of thousands or fewer) but positive rates between 1% and 20%; hours 8–19 carry 270,000–620,000 rows each with rates of 0.03%–0.1%; rates rise again after hour 20. Because positives are injected uniformly across the day while normal volume follows a daytime schedule, `step_hour_of_day` is informative, but the mechanism is the simulator's clock rather than behaviour, and this must be said in the report.

Destination prior count



Causal aggregate: earlier transactions to the same destination. Training sample.
Synthetic PaySim data; educational decision-support prototype; no AML determination.

Figure: eda_11_dest_prior_count.png

Observation: Positives more often target a destination with zero prior transactions (density about 5.4 versus 3.7 for normals) and are under-represented at higher prior counts. The destination aggregates carry modest signal in the expected direction; the origin aggregates do not (see eda_07) and are flagged for removal in feature selection (validation task V10).

Feature Selection

Scope

Feature set `primary`, training split only, seeded stratified subsample of 1,000,000 rows with 766 positives (research R-08). Fit scope: MI ['train'], L1 ['train'].

Before

24 columns: `type_CASH_IN`, `type_CASH_OUT`, `type_DEBIT`, `type_PAYMENT`, `type_TRANSFER`, `amount_bucket`, `log_amount`, `log_oldbalance_org`, `log_oldbalance_dest`, `amount_to_orig_balance_ratio`, `orig_zero_balance_flag`, `dest_zero_balance_flag`, `zero_amount_flag`, `dest_is_merchant`, `step_hour_of_day`, `orig_balance_delta`, `dest_balance_delta`, `orig_balance_inconsistent_flag`, `dest_balance_inconsistent_flag`, `orig_zero_after_flag`, `orig_prior_txn_count`, `orig_prior_amount_sum`, `dest_prior_txn_count`, `dest_prior_amount_sum`

Constant columns (no information on the subsample): none

Filter method: mutual information (top 12)

column	MI score	selected
<code>dest_balance_inconsistent_flag</code>	0.1872	yes
<code>orig_balance_inconsistent_flag</code>	0.1831	yes
<code>dest_zero_balance_flag</code>	0.1031	yes
<code>type_CASH_OUT</code>	0.0760	yes
<code>dest_is_merchant</code>	0.0711	yes
<code>type_PAYMENT</code>	0.0708	yes
<code>orig_zero_balance_flag</code>	0.0690	yes
<code>orig_zero_after_flag</code>	0.0379	yes
<code>type_CASH_IN</code>	0.0319	yes
<code>amount_bucket</code>	0.0247	yes
<code>step_hour_of_day</code>	0.0189	yes
<code>amount_to_orig_balance_ratio</code>	0.0059	yes
<code>type_TRANSFER</code>	0.0048	
<code>orig_balance_delta</code>	0.0032	
<code>dest_prior_txn_count</code>	0.0028	
<code>log_amount</code>	0.0008	
<code>log_oldbalance_org</code>	0.0008	
<code>dest_balance_delta</code>	0.0004	
<code>type_DEBIT</code>	0.0001	
<code>log_oldbalance_dest</code>	0.0001	
<code>dest_prior_amount_sum</code>	0.0001	
<code>orig_prior_txn_count</code>	0.0000	
<code>orig_prior_amount_sum</code>	0.0000	
<code>zero_amount_flag</code>	0.0000	

Embedded method: L1 logistic regression (C = 0.1, standardised inputs, balanced class weight)

column	coefficient	non-zero
<code>orig_zero_balance_flag</code>	8.8216	yes
<code>orig_balance_inconsistent_flag</code>	-5.6202	yes
<code>orig_zero_after_flag</code>	5.2283	yes

column	coefficient	non-zero
log_oldbalance_dest	-4.3553	yes
type_CASH_OUT	4.0187	yes
dest_balance_delta	-3.5508	yes
log_oldbalance_org	3.2417	yes
type_TRANSFER	2.8905	yes
dest_zero_balance_flag	-2.7713	yes
type_CASH_IN	-2.6181	yes
amount_to_orig_balance_ratio	2.2877	yes
step_hour_of_day	-1.0616	yes
log_amount	1.0095	yes
dest_balance_inconsistent_flag	0.7384	yes
amount_bucket	-0.6153	yes
orig_balance_delta	0.5111	yes
dest_prior_txn_count	-0.1582	yes
orig_prior_txn_count	-0.0329	yes
zero_amount_flag	0.0096	yes
dest_is_merchant	0.0000	
dest_prior_amount_sum	0.0000	
orig_prior_amount_sum	0.0000	
type_DEBIT	0.0000	
type_PAYMENT	0.0000	

After

Intersection (10): type_CASH_IN, type_CASH_OUT, amount_bucket, amount_to_orig_balance_ratio, orig_zero_balance_flag, dest_zero_balance_flag, step_hour_of_day, orig_balance_inconsistent_flag, dest_balance_inconsistent_flag, orig_zero_after_flag

Combine rule intersection_or_union_if_lt → **intersection (10 >= min_size 6)**.

Selected columns (10): type_CASH_IN, type_CASH_OUT, amount_bucket, amount_to_orig_balance_ratio, orig_zero_balance_flag, dest_zero_balance_flag, step_hour_of_day, orig_balance_inconsistent_flag, dest_balance_inconsistent_flag, orig_zero_after_flag

Dropped columns (14): type_DEBIT, type_PAYMENT, type_TRANSFER, log_amount, log_oldbalance_org, log_oldbalance_dest, zero_amount_flag, dest_is_merchant, orig_balance_delta, dest_balance_delta, orig_prior_txn_count, orig_prior_amount_sum, dest_prior_txn_count, dest_prior_amount_sum

Registry selected set (9 features): type_onehot, amount_bucket, amount_to_orig_balance_ratio, orig_zero_balance_flag, dest_zero_balance_flag, step_hour_of_day, orig_balance_inconsistent_flag, dest_balance_inconsistent_flag, orig_zero_after_flag

Why this combined set (task T041, written 2026-09-05 after reviewing the tables above)

Both methods were fitted on the same seeded stratified subsample of 1,000,000 training rows (about 770 positives). Their intersection has 10 columns, above the min_size of 6, so the intersection rule applied and no union fallback was needed. At registry level that is 9 features, because type_onehot is kept whole: a one-hot block cannot be partially selected, so the selected matrix carries all five type columns (13 columns in total).

The two methods agree on the account-state signals seen in EDA: orig_zero_balance_flag, orig_zero_after_flag, amount_to_orig_balance_ratio, dest_zero_balance_flag, the two balance-inconsistency flags, amount_bucket, step_hour_of_day, and the CASH_IN / CASH_OUT type indicators.

Two readings of the tables matter for later milestones:

- **Mutual information ranks the two balance-inconsistency flags first (0.187 and 0.183)**, ahead of transaction type. Those flags are simulator artifacts (DQ-05, eda_08). The `selected` set therefore contains four of the five batch-only features and is not a prediction-time-safe set. It is a statistical selection, not a deployment recommendation; `strict_pretx` remains the honest comparison in Milestone 5.
- **L1 coefficient signs are not individually interpretable.** `orig_zero_balance_flag` receives the largest coefficient (+8.8) although EDA shows zero-balance origins almost never positive. The feature is strongly collinear with `amount_to_orig_balance_ratio` ($\rho = 0.81$) and `log_oldbalance_org` ($\rho = -0.83$), so the signs offset each other. Only non-zero magnitude was used for selection.

Features dropped and why

Dropped column	Reason from the tables
<code>log_amount</code>	Redundant with <code>amount_bucket</code> ($\rho = 0.99$); MI prefers the training-fitted deciles (0.025 vs 0.001). L1 kept it, so it fails the intersection only.
<code>log_oldbalance_org</code> , <code>log_oldbalance_dest</code>	Large L1 coefficients but MI below 0.001; their information is carried by the zero-balance flags and the ratio.
<code>dest_is_merchant</code>	MI 0.071 (informative) but L1 coefficient exactly zero because <code>type_PAYMENT</code> and <code>dest_zero_balance_flag</code> ($\rho = 0.81$) encode the same rows.
<code>zero_amount_flag</code>	MI 0.000: only 4 training rows carry the flag, too few for a density estimate; L1 coefficient 0.01. Kept in the <code>primary</code> set as a documented artifact (DQ-03).
<code>orig_prior_txn_count</code> , <code>orig_prior_amount_sum</code>	MI 0.000 and near-zero coefficients: origins do not repeat (DQ-11, eda_07). Confirms the EDA recommendation to drop.
<code>dest_prior_txn_count</code> , <code>dest_prior_amount_sum</code>	Weak on both methods ($MI \leq 0.003$; $L1 \leq 0.16$ in magnitude).
<code>orig_balance_delta</code> , <code>dest_balance_delta</code>	Their information is captured by the inconsistency and zero-after flags; $MI \leq 0.003$.
<code>type_TRANSFER</code> , <code>type_PAYMENT</code> , <code>type_DEBIT</code> (columns)	Dropped at column level (one-hot redundancy: with two type columns present the others are implied) but retained in the matrix because <code>type_onehot</code> is one registry feature.

Caveats

- MI was estimated on a subsample with about 770 positives; the ranking of features below 0.01 is noisy and should not be over-read.
- The `selected` set will be evaluated as one candidate feature set in Milestone 5 alongside `primary`, `strict_pretx` and `posttx_ablation`; selection here makes no claim about model performance.
- Fit scope was training-only for both selectors (`fitted_on: ['train']`), enforced by the fit-scope recorder and `tests/test_features.py`.

PCA Report

Role

Configured role: **diagnostic_and_visualization**. Components are a diagnostic of feature redundancy and a visualisation aid. They do not enter the primary model candidates. A `pca_variant` matrix (9 components plus type

one-hot) is written for one documented experiment in Milestone 5.

Inputs

12 standardised numeric/aggregate training features: amount_bucket, log_amount, log_oldbalance_org, log_oldbalance_dest, amount_to_orig_balance_ratio, step_hour_of_day, orig_balance_delta, dest_balance_delta, orig_prior_txn_count, orig_prior_amount_sum, dest_prior_txn_count, dest_prior_amount_sum

Fit scope: ['train'] (scaler and PCA fitted on training rows only).

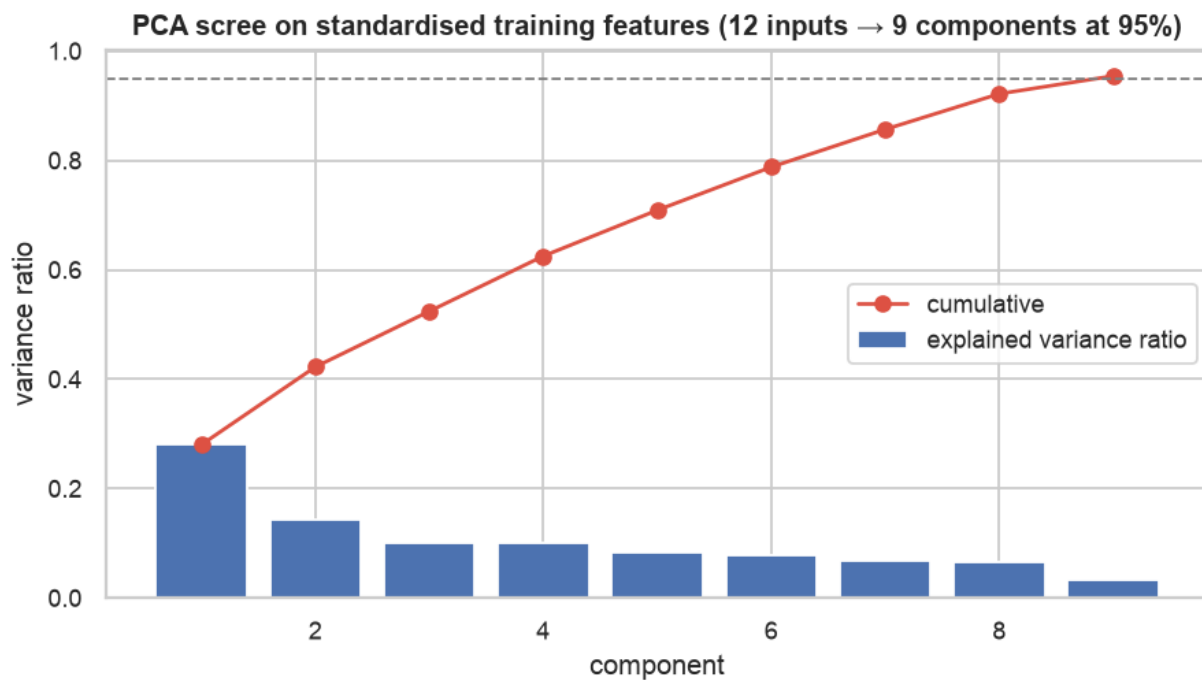
Explained variance (9 components reach the 0.95 target)

component	variance ratio	cumulative
PC1	0.2803	0.2803
PC2	0.1423	0.4226
PC3	0.1017	0.5243
PC4	0.1009	0.6253
PC5	0.0839	0.7092
PC6	0.0789	0.7881
PC7	0.0689	0.8570
PC8	0.0650	0.9220
PC9	0.0327	0.9546

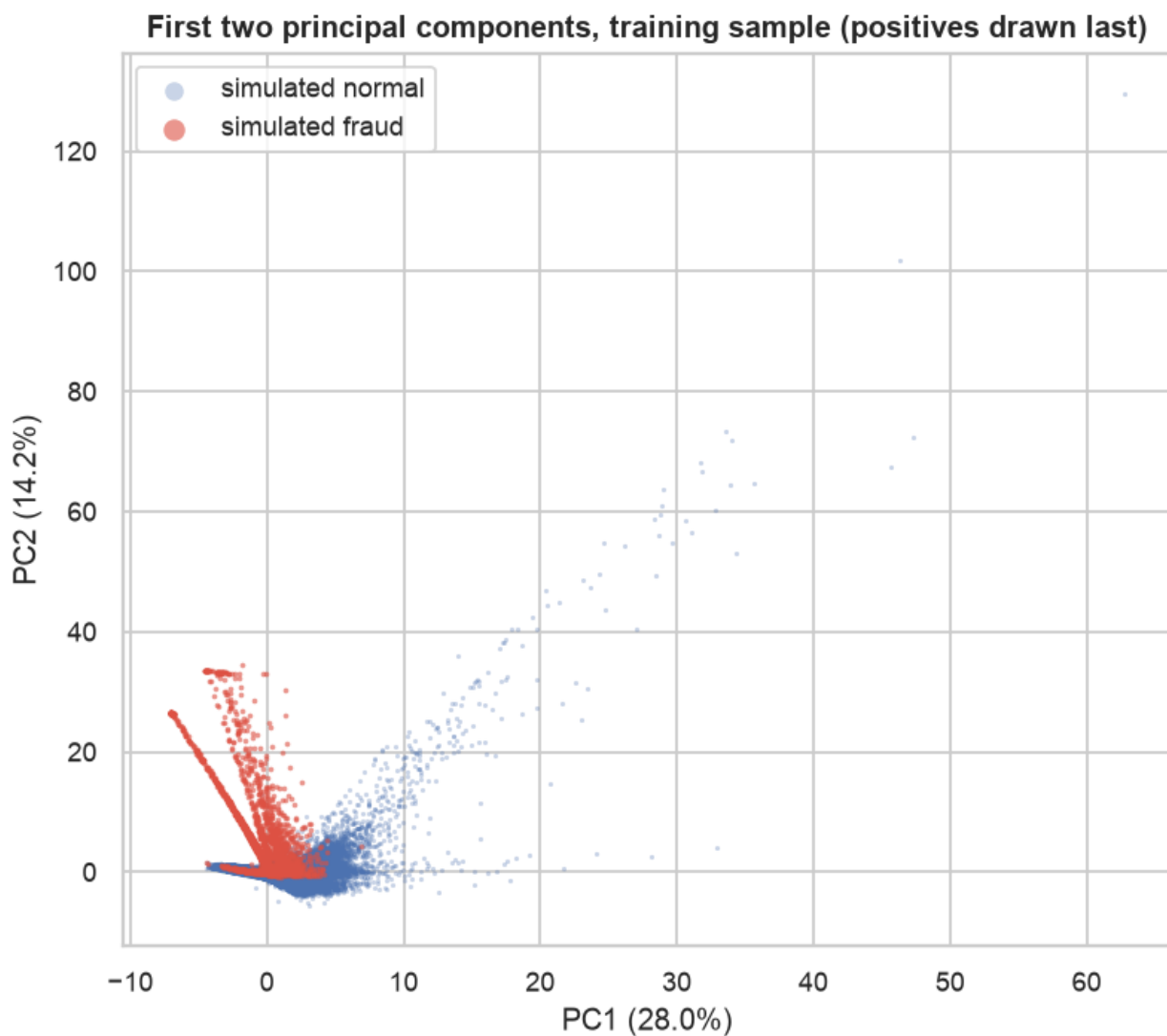
Top loadings per component

component	largest absolute loadings
PC1	log_amount (+0.46), amount_bucket (+0.45), log_oldbalance_dest (+0.44), dest_prior_txn_count (+0.40)
PC2	dest_balance_delta (+0.59), amount_to_orig_balance_ratio (+0.57), orig_balance_delta (+0.36), log_oldbalance_org (-0.34)
PC3	orig_prior_txn_count (+0.69), orig_prior_amount_sum (+0.69), dest_prior_amount_sum (+0.10), dest_prior_txn_count (+0.09)
PC4	dest_prior_amount_sum (+0.53), dest_prior_txn_count (+0.48), log_amount (-0.38), amount_bucket (-0.37)
PC5	step_hour_of_day (+0.66), log_oldbalance_org (+0.49), orig_balance_delta (-0.43), amount_to_orig_balance_ratio (+0.27)
PC6	step_hour_of_day (+0.69), log_oldbalance_org (-0.40), orig_balance_delta (+0.33), dest_prior_amount_sum (-0.25)
PC7	orig_balance_delta (+0.72), log_oldbalance_org (+0.65), amount_to_orig_balance_ratio (-0.16), dest_balance_delta (+0.12)
PC8	orig_prior_txn_count (+0.71), orig_prior_amount_sum (-0.71), log_oldbalance_org (-0.00), step_hour_of_day (-0.00)
PC9	log_oldbalance_dest (+0.80), dest_prior_amount_sum (-0.44), amount_bucket (-0.27), log_amount (-0.26)

Figures



Fitted on the training split only; inputs are numeric and aggregate features (flags and one-hots excluded).
Synthetic PaySim data; educational decision-support prototype; no AML determination.



Seeded sample: all training positives plus sampled negatives; projection uses the training-fitted scaler and PCA.
Synthetic PaySim data; educational decision-support prototype; no AML determination.

Interpretation (task T041, written 2026-09-05 after reviewing the tables and figures above)

Nine components are needed to reach 95% of the variance of the 12 standardised numeric and aggregate features, and the first two explain only 28.0% and 14.2%. Variance is spread evenly, so the numeric block is not highly redundant apart from three known pairs: `log_amount/amount_bucket` (both load +0.45 on PC1), the two destination aggregates (PC4), and the two origin aggregates, which form components PC3 and PC8 on their own. Those origin aggregates are near-constant in raw units; standardisation inflates them to unit variance, which is why they earn two components while carrying no information (eda_07). This is a diagnostic confirmation that they belong out of the model.

In the projection (`pca_02_projection.png`) positives concentrate along a narrow ray with negative PC1 and PC2 rising to about 30. PC2 loads on `dest_balance_delta` (+0.59), `amount_to_orig_balance_ratio` (+0.57) and `orig_balance_delta` (+0.36) against `log_oldbalance_org` (−0.34): it is the "amount moves the whole origin balance to the destination" direction seen in eda_09. Normals spread along positive PC1 (large amounts, funded destinations with prior activity). The classes overlap near the origin, so two components separate only the extreme positives.

Do components enter any model?

No. The configured role is diagnostic and visualisation. The primary candidates use raw engineered features because:

- the artifact ablation (research R-06) needs artifact-driven and behavioural features kept separate, and every component mixes both (PC2 blends the balance deltas with the ratio);
- SHAP and PDP explanations on raw features are readable by investigators; explanations on components are not;
- tree ensembles gain nothing from an orthogonal rotation of inputs.

A `pca_variant` matrix (9 components plus the type one-hot; fit scope `['train']`) is written so Milestone 5 can run one documented experiment and report whether the rotation costs or gains anything. No performance claim is made here.

4. Model Comparison and Selection

Model Comparison

Method

Every candidate is trained on the training split of its feature set and scored on the split named in each section; comparators need no training. Review period = 24 steps; primary K = 200; k_grid = [50, 100, 200, 300, 500]. Threshold metrics use 0.5 until the operating point is chosen on validation (Milestone 6). Accuracy appears last, next to the majority-class baseline, and is never a selection criterion (FR-007). PR-AUC is primary; the no-skill PR-AUC equals prevalence.

Validation: headline metrics

candidate [feature set]	PR-AUC	ROC-AUC	Recall@200 (mean/period)	Recall@200 (pooled)	Precision@200 (mean/period)	Brier	ECE	degenerate
balanced_rf [primary]	1.0000	1.0000	0.8029	0.7979	1.0000	0.0004	0.0046	
hgb [posttx_ablation]	1.0000	1.0000	0.8029	0.7979	1.0000	0.0007	0.0024	
hgb [primary]	1.0000	1.0000	0.8029	0.7979	1.0000	0.0000	0.0000	
hgb [strict_pretx]	1.0000	1.0000	0.8029	0.7979	1.0000	0.0000	0.0000	
balanced_rf [strict_pretx]	1.0000	1.0000	0.8029	0.7979	1.0000	0.0008	0.0048	
hgb [selected]	0.9990	0.9994	0.8029	0.7979	1.0000	0.0003	0.0006	
logreg [primary]	0.9987	0.9992	0.8029	0.7979	1.0000	0.0003	0.0065	
hgb [pca_variant]	0.9069	0.9972	0.7526	0.7500	0.9400	0.0070	0.0113	
logreg [strict_pretx]	0.2776	0.9824	0.3117	0.3145	0.3942	0.0589	0.0893	
rule comparator (flag, then amount)	0.1555	0.8142	0.1965	0.1995	0.2500	0.0077	0.0024	
random ranking	0.0084	0.5050	0.0104	0.0106	0.0133	0.3335	0.4919	
dummy (chronological order) [primary]	0.0083	0.5000	0.0650	0.0665	0.0833	0.0083	0.0075	yes
dummy [strict_pretx]	0.0083	0.5000	0.0650	0.0665	0.0833	0.0083	0.0075	yes

Validation: Recall@K across the capacity grid (mean over review periods)

candidate [feature set]	Recall@50	Recall@100	Recall@200	Recall@300	Recall@500
balanced_rf [primary]	0.2007	0.4015	0.8029	1.0000	1.0000
hgb [posttx_ablation]	0.2007	0.4015	0.8029	1.0000	1.0000
hgb [primary]	0.2007	0.4015	0.8029	1.0000	1.0000
hgb [strict_pretx]	0.2007	0.4015	0.8029	1.0000	1.0000
balanced_rf [strict_pretx]	0.2007	0.4015	0.8029	1.0000	1.0000
hgb [selected]	0.2007	0.4015	0.8029	0.9987	0.9993

candidate [feature set]	Recall@50	Recall@100	Recall@200	Recall@300	Recall@500
logreg [primary]	0.2007	0.4015	0.8029	0.9988	0.9988
hgb [pca_variant]	0.2007	0.4007	0.7526	0.8720	0.9300
logreg [strict_pretx]	0.0719	0.1617	0.3117	0.4220	0.5787
rule comparator (flag, then amount)	0.0927	0.1352	0.1965	0.2351	0.3102
random ranking	0.0013	0.0026	0.0104	0.0130	0.0220
dummy (chronological order) [primary]	0.0457	0.0528	0.0650	0.0925	0.1513
dummy [strict_pretx]	0.0457	0.0528	0.0650	0.0925	0.1513

Validation: Precision@K across the capacity grid (mean over review periods)

candidate [feature set]	Precision@50	Precision@100	Precision@200	Precision@300	Precision@500
balanced_rf [primary]	1.0000	1.0000	1.0000	0.8356	0.5013
hgb [posttx_ablation]	1.0000	1.0000	1.0000	0.8356	0.5013
hgb [primary]	1.0000	1.0000	1.0000	0.8356	0.5013
hgb [strict_pretx]	1.0000	1.0000	1.0000	0.8356	0.5013
balanced_rf [strict_pretx]	1.0000	1.0000	1.0000	0.8356	0.5013
hgb [selected]	1.0000	1.0000	1.0000	0.8344	0.5010
logreg [primary]	1.0000	1.0000	1.0000	0.8344	0.5007
hgb [pca_variant]	1.0000	0.9983	0.9400	0.7289	0.4670
logreg [strict_pretx]	0.3667	0.4100	0.3942	0.3556	0.2923
rule comparator (flag, then amount)	0.4733	0.3433	0.2500	0.1989	0.1567
random ranking	0.0067	0.0067	0.0133	0.0111	0.0113
dummy (chronological order) [primary]	0.2333	0.1350	0.0833	0.0789	0.0770
dummy [strict_pretx]	0.2333	0.1350	0.0833	0.0789	0.0770

Validation: threshold metrics at 0.5

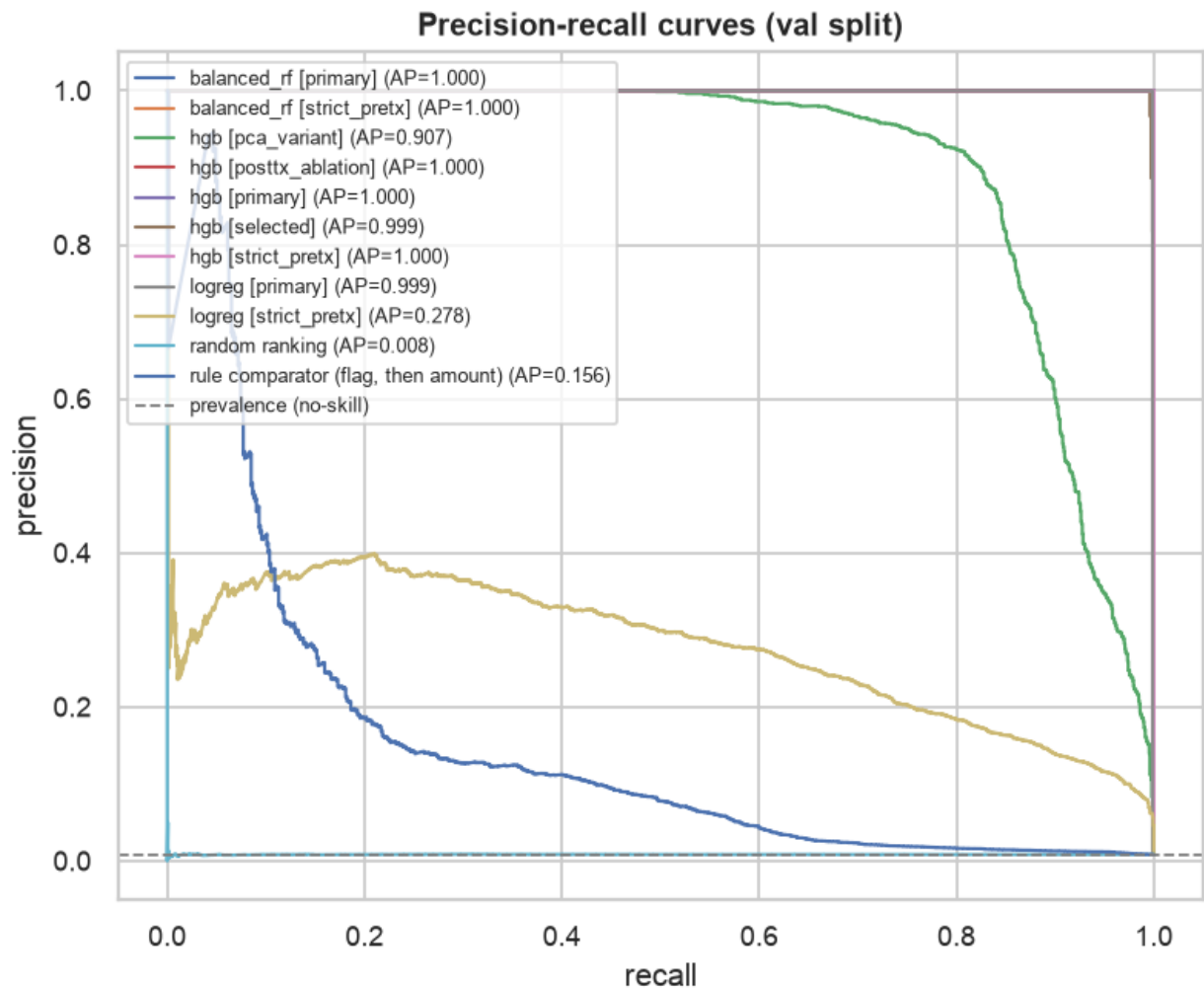
candidate [feature set]	threshold	precision	recall	F1	FPR	TP	FP	FN	TN
balanced_rf [primary]	0.5000	0.9967	1.0000	0.9983	0.0000	1,504	5	0	179,559
hgb [posttx_ablation]	0.5000	0.9477	1.0000	0.9731	0.0005	1,504	83	0	179,481
hgb [primary]	0.5000	1.0000	1.0000	1.0000	0.0000	1,504	0	0	179,564
hgb [strict_pretx]	0.5000	1.0000	0.9993	0.9997	0.0000	1,503	0	1	179,564
balanced_rf [strict_pretx]	0.5000	0.9284	1.0000	0.9629	0.0006	1,504	116	0	179,448
hgb [selected]	0.5000	0.9671	0.9973	0.9820	0.0003	1,500	51	4	179,513
logreg [primary]	0.5000	0.9967	0.9987	0.9977	0.0000	1,502	5	2	179,559

candidate [feature set]	threshold	precision	recall	F1	FPR	TP	FP	FN	TN
hgb [pca_variant]	0.5000	0.4764	0.9242	0.6287	0.0085	1,390	1,528	114	178,036
logreg [strict_pretx]	0.5000	0.0906	0.9820	0.1658	0.0826	1,477	14,834	27	164,730
rule comparator (flag, then amount)	0.5000	1.0000	0.0013	0.0027	0.0000	2	0	1,502	179,564
random ranking	0.5000	0.0084	0.5073	0.0166	0.5006	763	89,883	741	89,681
dummy (chronological order) [primary]	0.5000	0.0000	0.0000	0.0000	0.0000	0	0	1,504	179,564
dummy [strict_pretx]	0.5000	0.0000	0.0000	0.0000	0.0000	0	0	1,504	179,564

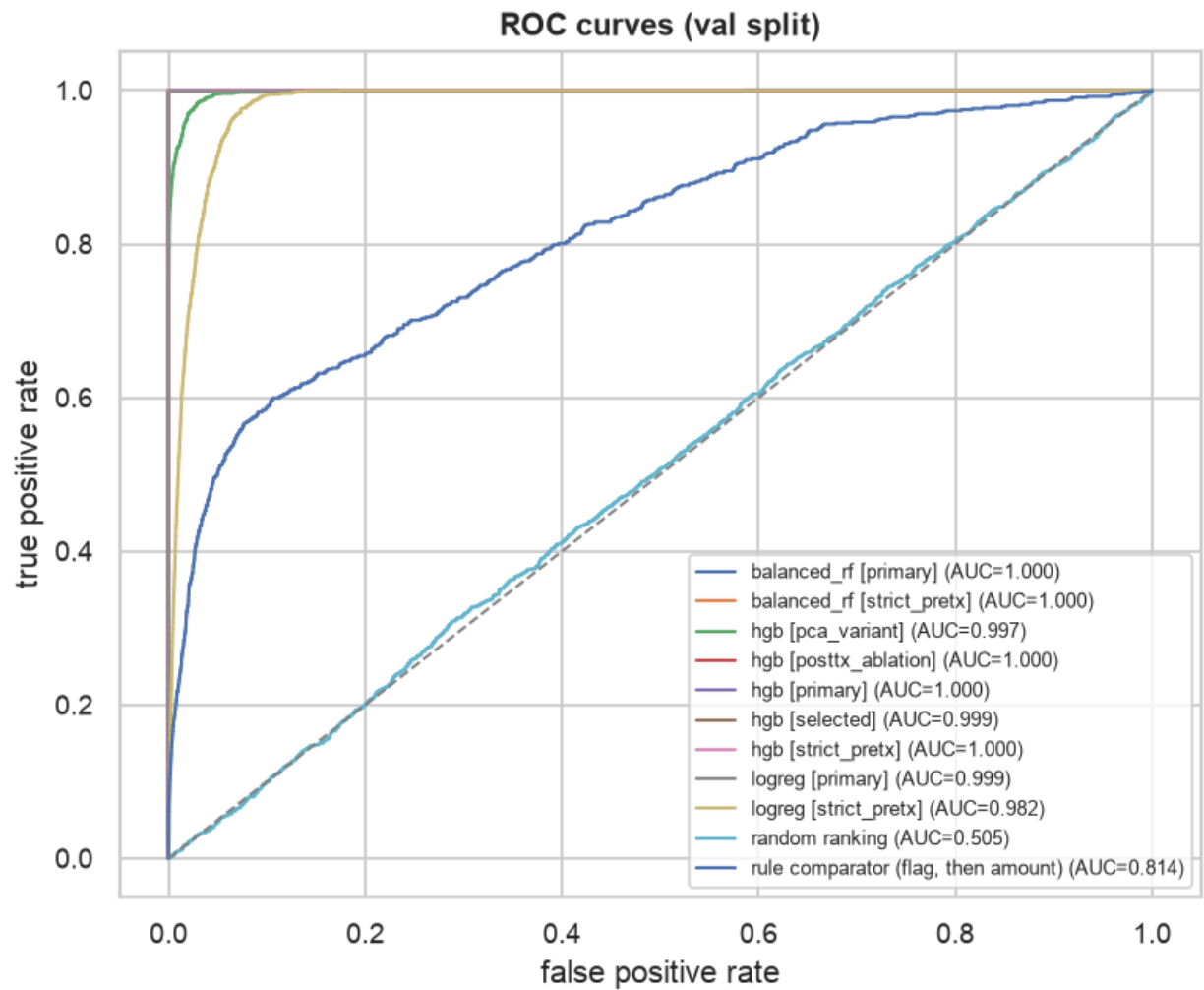
Validation: accuracy (reported last, with prevalence)

candidate [feature set]	accuracy	prevalence	majority-class accuracy (1 - prevalence)
balanced_rf [primary]	1.0000	0.0083	0.9917
hgb [posttx_ablation]	0.9995	0.0083	0.9917
hgb [primary]	1.0000	0.0083	0.9917
hgb [strict_pretx]	1.0000	0.0083	0.9917
balanced_rf [strict_pretx]	0.9994	0.0083	0.9917
hgb [selected]	0.9997	0.0083	0.9917
logreg [primary]	1.0000	0.0083	0.9917
hgb [pca_variant]	0.9909	0.0083	0.9917
logreg [strict_pretx]	0.9179	0.0083	0.9917
rule comparator (flag, then amount)	0.9917	0.0083	0.9917
random ranking	0.4995	0.0083	0.9917
dummy (chronological order) [primary]	0.9917	0.0083	0.9917
dummy [strict_pretx]	0.9917	0.0083	0.9917

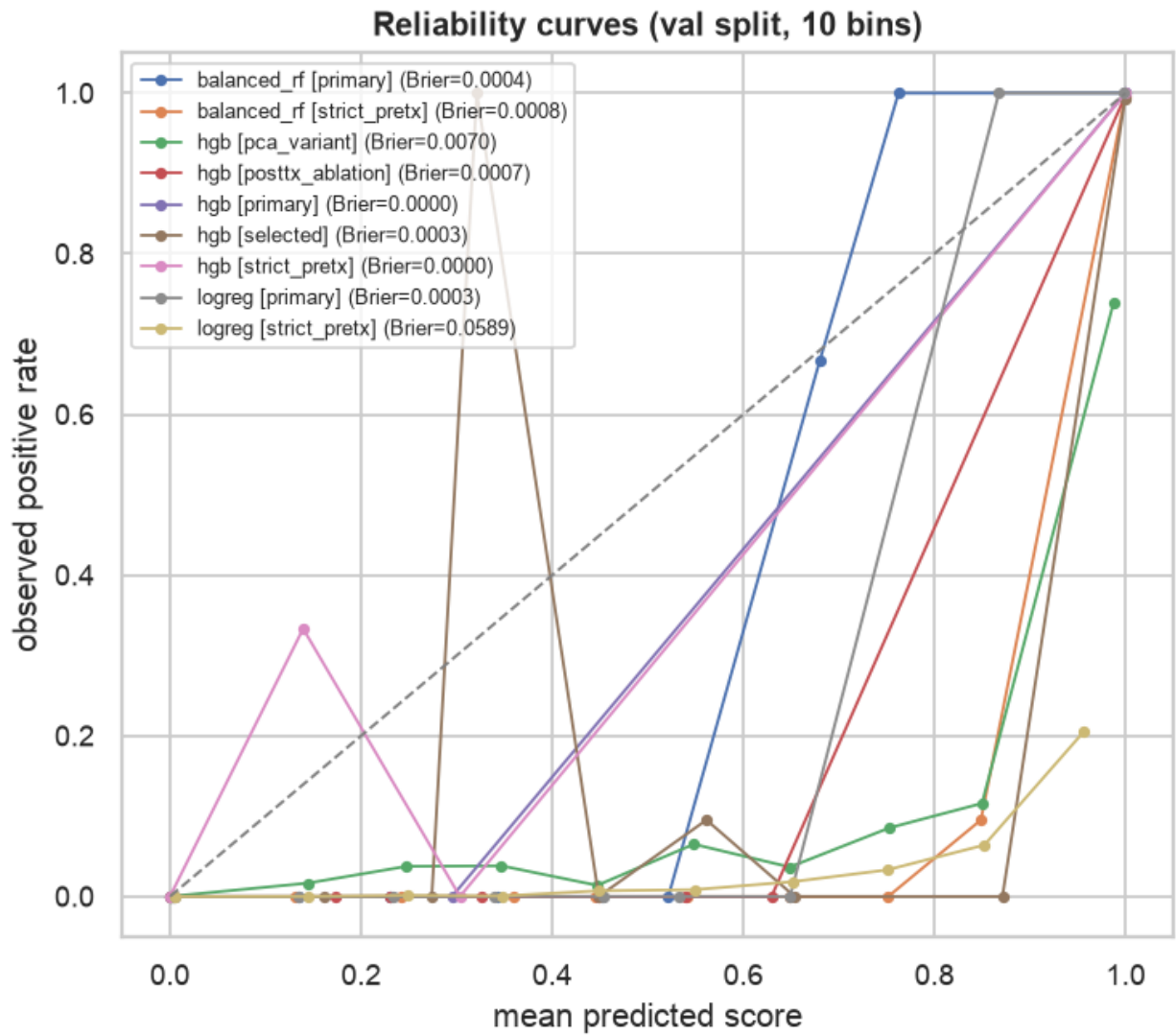
Validation: curves



Degenerate (constant-score) candidates omitted; the dashed line is the no-skill PR-AUC (prevalence).
 Synthetic PaySim data; educational decision-support prototype; no AML determination.



ROC is reported as a secondary metric; PR-AUC is primary under 0.1-1% prevalence.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.



Class-weighted models are expected to over-predict; calibration is assessed on validation before any correction (research R-09).
 Synthetic PaySim data; educational decision-support prototype; no AML determination.

Test (single-touch evaluation): headline metrics

candidate [feature set]	PR-AUC	ROC-AUC	Recall@200 (mean/period)	Recall@200 (pooled)	Precision@200 (mean/period)	Brier	ECE	degenerate
hgb [primary]	1.0000	1.0000	0.7568	0.7547	1.0000	0.0000	0.0000	
hgb [posttx_ablation]	1.0000	1.0000	0.7568	0.7547	1.0000	0.0007	0.0023	
balanced_rf [strict_pretx]	0.9997	1.0000	0.7568	0.7547	1.0000	0.0007	0.0043	
balanced_rf [primary]	0.9997	1.0000	0.7568	0.7547	1.0000	0.0003	0.0038	
hgb [selected]	0.9996	1.0000	0.7568	0.7547	1.0000	0.0004	0.0007	
hgb [strict_pretx]	0.9995	0.9995	0.7568	0.7547	1.0000	0.0000	0.0000	
logreg [primary]	0.9954	0.9971	0.7568	0.7547	1.0000	0.0004	0.0070	
hgb [pca_variant]	0.9025	0.9971	0.7313	0.7302	0.9675	0.0080	0.0127	
logreg [strict_pretx]	0.2908	0.9788	0.3539	0.3575	0.4738	0.0687	0.1022	

candidate [feature set]	PR-AUC	ROC-AUC	Recall@200 (mean/period)	Recall@200 (pooled)	Precision@200 (mean/period)	Brier	ECE	degenerate
rule comparator (flag, then amount)	0.1856	0.7998	0.3101	0.3142	0.4163	0.0101	0.0054	
random ranking	0.0109	0.4990	0.1012	0.1038	0.1375	0.3334	0.4891	
dummy (chronological order) [primary]	0.0109	0.5000	0.2076	0.2132	0.2825	0.0109	0.0102	yes
dummy [strict_pretx]	0.0109	0.5000	0.2076	0.2132	0.2825	0.0109	0.0102	yes

Test (single-touch evaluation): Recall@K across the capacity grid (mean over review periods)

candidate [feature set]	Recall@50	Recall@100	Recall@200	Recall@300	Recall@500
hgb [primary]	0.1892	0.3784	0.7568	1.0000	1.0000
hgb [posttx_ablation]	0.1892	0.3784	0.7568	1.0000	1.0000
balanced_rf [strict_pretx]	0.1892	0.3784	0.7568	0.9996	1.0000
balanced_rf [primary]	0.1892	0.3784	0.7568	0.9990	0.9995
hgb [selected]	0.1892	0.3784	0.7568	0.9995	1.0000
hgb [strict_pretx]	0.1892	0.3784	0.7568	0.9995	0.9995
logreg [primary]	0.1892	0.3784	0.7568	0.9967	0.9967
hgb [pca_variant]	0.1892	0.3784	0.7313	0.8815	0.9384
logreg [strict_pretx]	0.0876	0.1781	0.3539	0.4835	0.6324
rule comparator (flag, then amount)	0.1119	0.1993	0.3101	0.3869	0.4451
random ranking	0.0258	0.0507	0.1012	0.1385	0.1476
dummy (chronological order) [primary]	0.0701	0.1099	0.2076	0.3188	0.3600
dummy [strict_pretx]	0.0701	0.1099	0.2076	0.3188	0.3600

Test (single-touch evaluation): Precision@K across the capacity grid (mean over review periods)

candidate [feature set]	Precision@50	Precision@100	Precision@200	Precision@300	Precision@500
hgb [primary]	1.0000	1.0000	1.0000	0.8950	0.5870
hgb [posttx_ablation]	1.0000	1.0000	1.0000	0.8950	0.5870
balanced_rf [strict_pretx]	1.0000	1.0000	1.0000	0.8946	0.5870
balanced_rf [primary]	1.0000	1.0000	1.0000	0.8942	0.5868
hgb [selected]	1.0000	1.0000	1.0000	0.8946	0.5870
hgb [strict_pretx]	1.0000	1.0000	1.0000	0.8946	0.5867
logreg [primary]	1.0000	1.0000	1.0000	0.8921	0.5853
hgb [pca_variant]	1.0000	1.0000	0.9675	0.7904	0.5547
logreg [strict_pretx]	0.4700	0.4775	0.4738	0.4425	0.3942
rule comparator (flag, then amount)	0.6000	0.5350	0.4163	0.3575	0.2953
random ranking	0.1400	0.1375	0.1375	0.1371	0.1373

candidate [feature set]	Precision@50	Precision@100	Precision@200	Precision@300	Precision@500
dummy (chronological order) [primary]	0.3850	0.3000	0.2825	0.3004	0.2512
dummy [strict_pretx]	0.3850	0.3000	0.2825	0.3004	0.2512

Test (single-touch evaluation): threshold metrics at 0.5

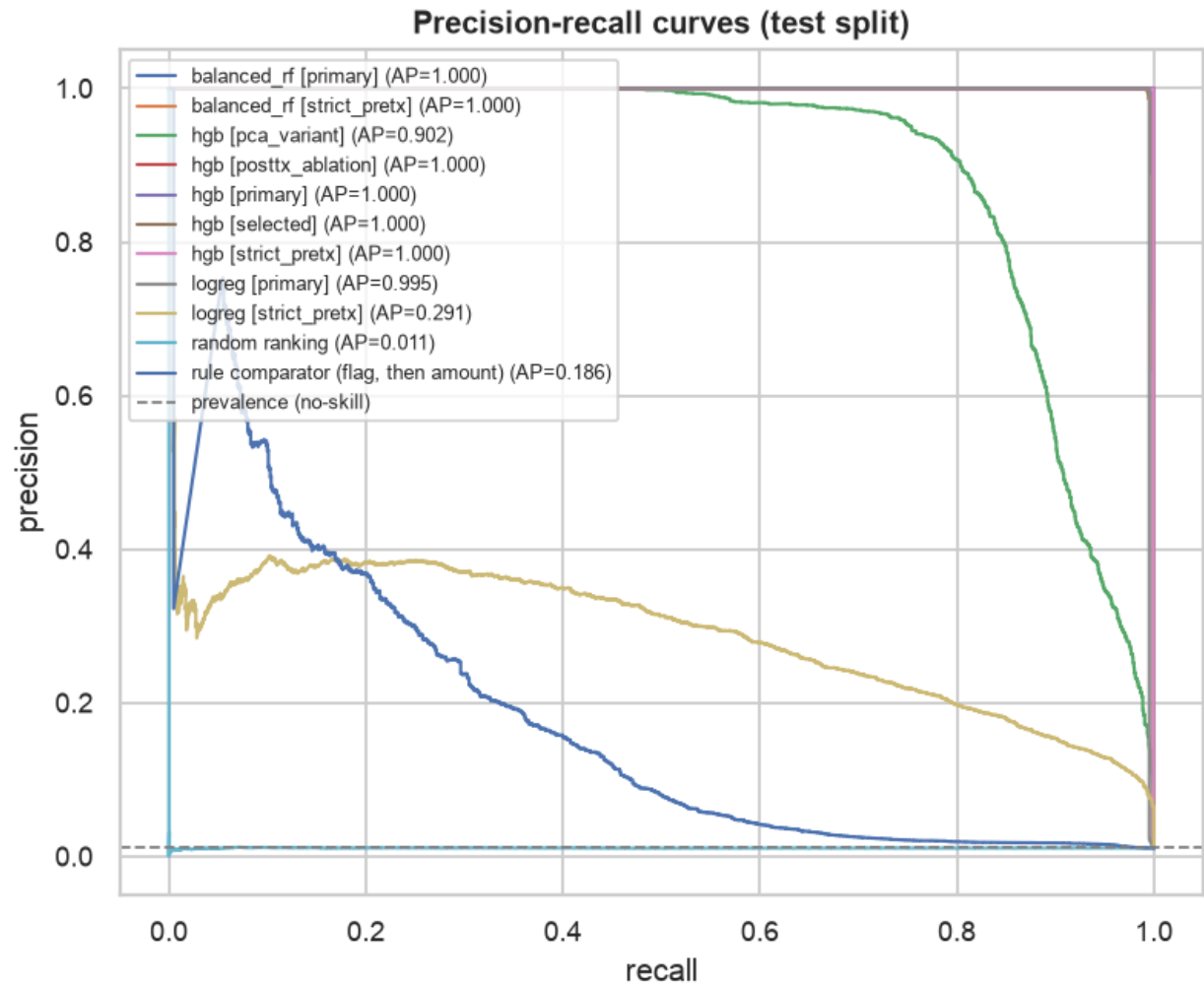
candidate [feature set]	threshold	precision	recall	F1	FPR	TP	FP	FN	TN
hgb [primary]	0.5000	1.0000	1.0000	1.0000	0.0000	2,120	0	0	192,015
hgb [posttx_ablation]	0.5000	0.9532	1.0000	0.9761	0.0005	2,120	104	0	191,911
balanced_rf [strict_pretx]	0.5000	0.9405	0.9995	0.9691	0.0007	2,119	134	1	191,881
balanced_rf [primary]	0.5000	0.9972	0.9991	0.9981	0.0000	2,118	6	2	192,009
hgb [selected]	0.5000	0.9751	0.9972	0.9860	0.0003	2,114	54	6	191,961
hgb [strict_pretx]	0.5000	1.0000	0.9991	0.9995	0.0000	2,118	0	2	192,015
logreg [primary]	0.5000	0.9943	0.9953	0.9948	0.0001	2,110	12	10	192,003
hgb [pca_variant]	0.5000	0.4994	0.9075	0.6442	0.0100	1,924	1,929	196	190,086
logreg [strict_pretx]	0.5000	0.1012	0.9844	0.1836	0.0965	2,087	18,529	33	173,486
rule comparator (flag, then amount)	0.5000	1.0000	0.0047	0.0094	0.0000	10	0	2,110	192,015
random ranking	0.5000	0.0107	0.4906	0.0210	0.5002	1,040	96,037	1,080	95,978
dummy (chronological order) [primary]	0.5000	0.0000	0.0000	0.0000	0.0000	0	0	2,120	192,015
dummy [strict_pretx]	0.5000	0.0000	0.0000	0.0000	0.0000	0	0	2,120	192,015

Test (single-touch evaluation): accuracy (reported last, with prevalence)

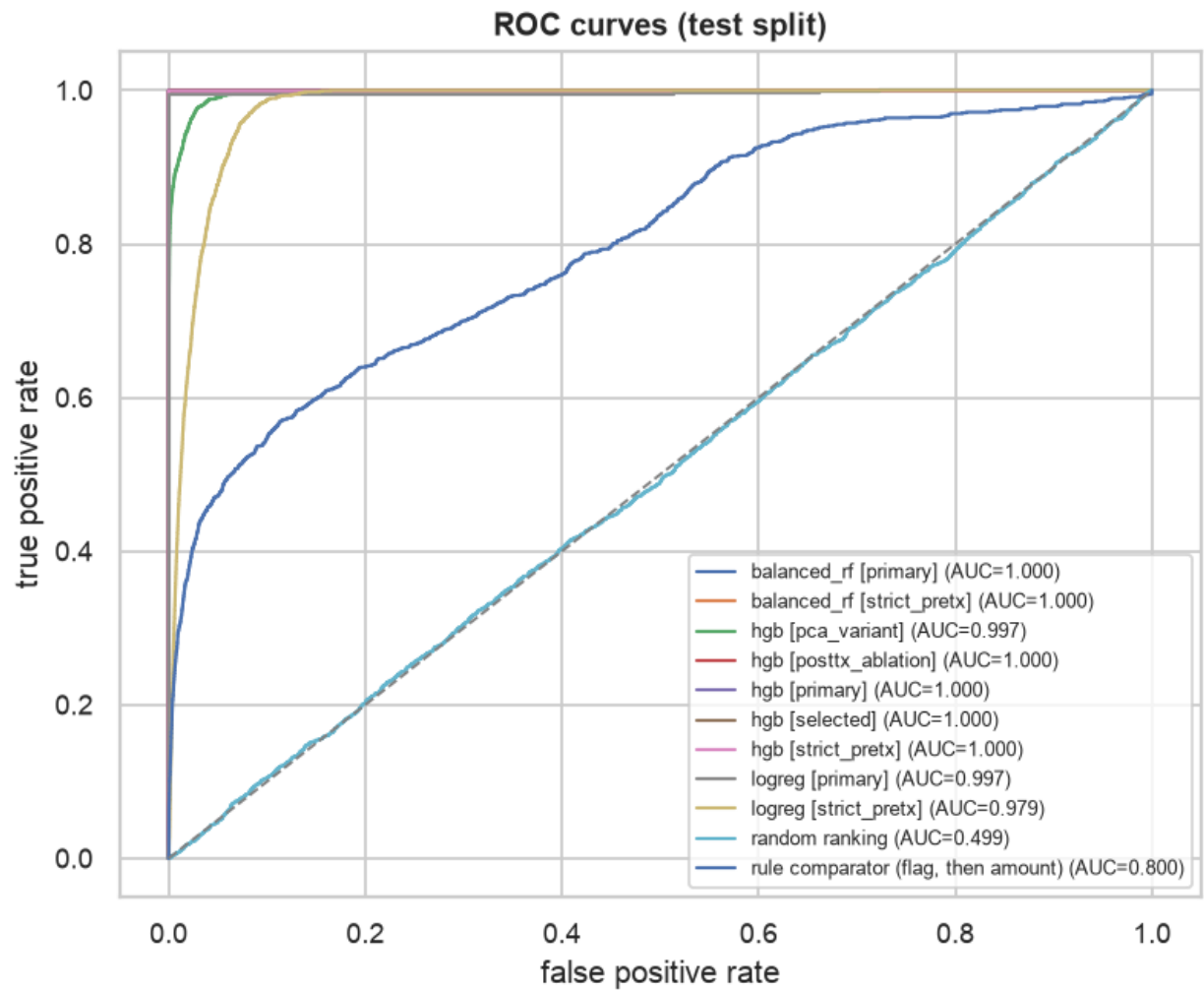
candidate [feature set]	accuracy	prevalence	majority-class accuracy (1 - prevalence)
hgb [primary]	1.0000	0.0109	0.9891
hgb [posttx_ablation]	0.9995	0.0109	0.9891
balanced_rf [strict_pretx]	0.9993	0.0109	0.9891
balanced_rf [primary]	1.0000	0.0109	0.9891
hgb [selected]	0.9997	0.0109	0.9891
hgb [strict_pretx]	1.0000	0.0109	0.9891
logreg [primary]	0.9999	0.0109	0.9891
hgb [pca_variant]	0.9891	0.0109	0.9891
logreg [strict_pretx]	0.9044	0.0109	0.9891

candidate [feature set]	accuracy	prevalence	majority-class accuracy (1 - prevalence)
rule comparator (flag, then amount)	0.9891	0.0109	0.9891
random ranking	0.4997	0.0109	0.9891
dummy (chronological order) [primary]	0.9891	0.0109	0.9891
dummy [strict_pretx]	0.9891	0.0109	0.9891

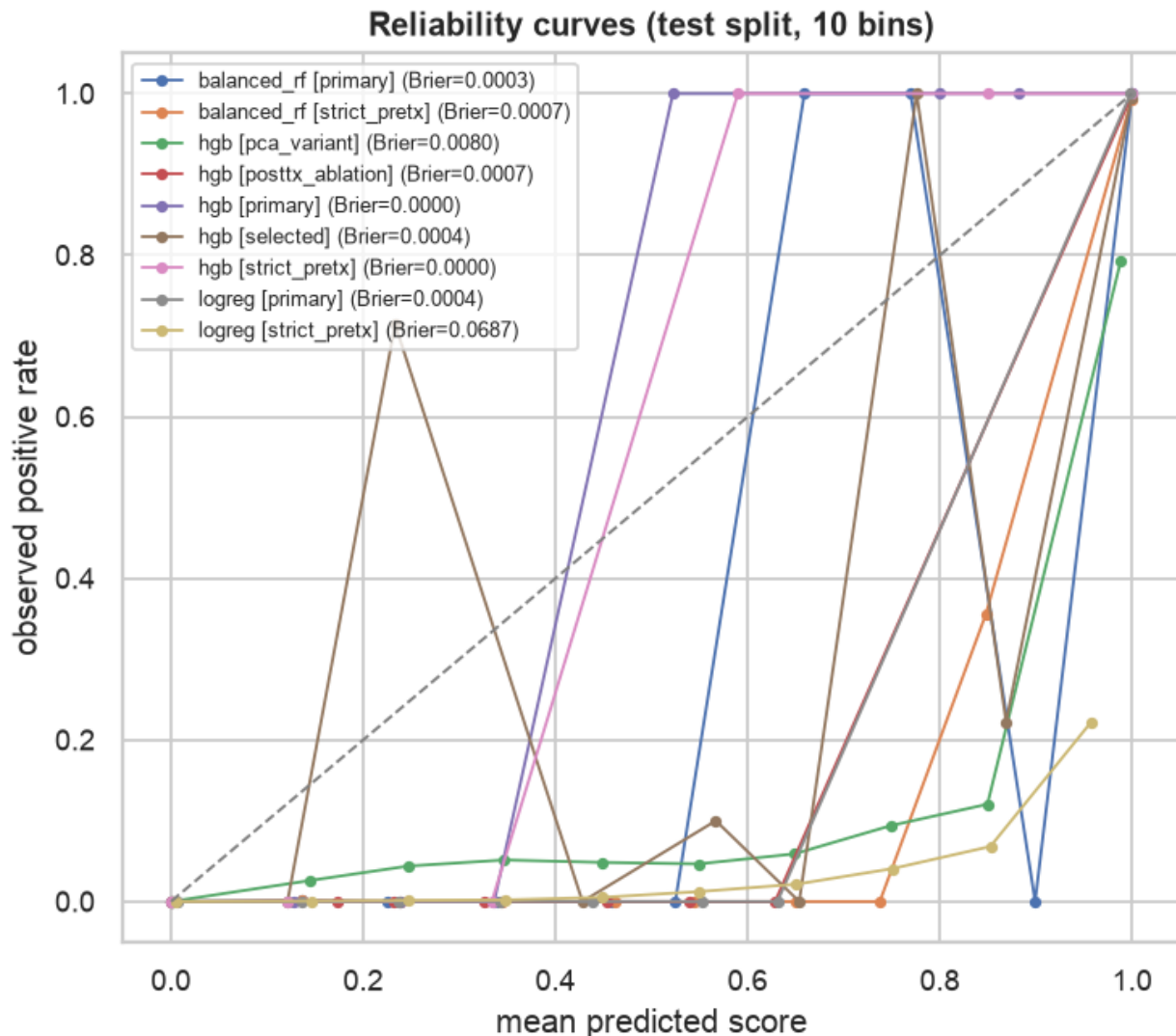
Test (single-touch evaluation): curves



Degenerate (constant-score) candidates omitted; the dashed line is the no-skill PR-AUC (prevalence).
 Synthetic PaySim data; educational decision-support prototype; no AML determination.



ROC is reported as a secondary metric; PR-AUC is primary under 0.1-1% prevalence.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.



Class-weighted models are expected to over-predict; calibration is assessed on validation before any correction (research R-09).
 Synthetic PaySim data; educational decision-support prototype; no AML determination.

Validation discussion (task T050, written 2026-09-05 after reviewing the tables and curves above)

All numbers below are validation-split figures (steps 409–552, 181,068 rows, 1,504 positives, six review periods). No test-split number appears here; the test split stays locked until the operating point is frozen (Milestone 6).

1. The simulator is almost perfectly separable, and that is the main finding

Every tree model on every non-PCA feature set reaches PR-AUC 1.0000 and ROC-AUC 1.0000: hgb on primary, strict_pretx, selected and posttx_ablation, and balanced_rf on primary (balanced_rf on strict_pretx is 0.9997). At the default 0.5 threshold, hgb [primary] makes one false positive and zero false negatives across 181,068 validation rows. Logistic regression on the primary set is close behind at 0.9987.

This is not evidence of AML capability. PaySim generates its positives from a small set of agent rules (an account is drained by TRANSFER and CASH_OUT), so a handful of engineered features reproduce the generator's own rule almost exactly: amount_to_orig_balance_ratio near 1, orig_zero_after_flag, and the balance-arithmetic flags (eda_08, eda_09). Any model that can express "ratio $\approx$ 1 and type $\in$ {TRANSFER, CASH_OUT}" recovers the label. The comparison therefore says a lot about the dataset and little about which learner would be best on real transactions.

2. The artifact ablation (research R-06) reads differently for trees and for the linear model

- For tree models the expected gap between primary and strict_pretx does **not** appear: hgb scores 1.0000 on both. The post-transaction artifact features are sufficient on their own (posttx_ablation, which contains only

type plus the five batch-only features, also reaches 1.0000) but they are not necessary: the pre-transaction interaction between type and the amount-to-balance ratio carries the same information for a non-linear learner.

- For logistic regression the gap is large: 0.9987 on `primary` versus 0.2599 on `strict_pretx`, with its `strict_pretx` PR curve peaking near 0.36 precision. A linear model cannot express the type-by-ratio interaction additively, so it needs the post-transaction flags to do the work.
- `hgb [pca_variant]` reaches 0.8998: rotating the numeric block into nine components mixes the ratio with unrelated variance and costs about 0.10 of PR-AUC and 0.07 of Recall@200. Components do not enter any further candidate (see `reports/pca_report.md`).

3. Recall@K is bounded by K on validation, not by the models

Validation periods hold between 216 and 272 positives, all above $K = 200$. The best possible mean Recall@200 is therefore the mean of $200 / \text{positives}$ over the six periods, which is 0.8029; every strong model hits exactly that ceiling with Precision@200 = 1.0000, meaning all 200 reviewed transactions in every period are positives. The per-period recall for `hgb [primary]` ranges from 0.735 (272 positives) to 0.926 (216 positives) purely because of the positive count. At $K = 300$ and $K = 500$ the strong models reach 1.0000 because K exceeds the positives; at $K = 50$ and $K = 100$ they all sit at $K / \text{positives}$ (0.2007 and 0.4015). Recall@ K cannot separate the strong candidates on this split; it does show that $K = 200$ binds, which is the intended operational reading.

4. Comparators

The rule comparator (flag, then amount) reaches PR-AUC 0.1555 and Recall@200 of 0.1965: amount alone puts about one positive in five into the daily top 200, and its threshold row shows only two flagged rows in the whole validation split (precision 1.0, recall 0.0013). Random ranking gives PR-AUC 0.0084, equal to prevalence (0.0083), and Recall@200 of 0.0104. The dummy candidate's constant scores rank as chronological order and give Recall@200 of 0.0650. Every learner except `logreg [strict_pretx]` beats all three comparators by a wide margin; `logreg [strict_pretx]` still beats them (0.2945 versus 0.1965 at $K = 200$).

5. Calibration

The reliability curves lie below the diagonal in the middle of the score range for every model: class weighting inflates mid-range scores, as expected. Because the strong models push almost every row to a score near 0 or 1, their Brier scores are tiny (`hgb [primary]` 0.0000, `logreg [primary]` 0.0003, `balanced_rf [primary]` 0.0005) and ECE is below 0.01 for all of them. `balanced_rf [primary]` is the least calibrated of the strong models (ECE 0.0090; its 0.55–0.70 bins show observed rates of 0.66–1.00). Whether isotonic calibration fitted on validation helps is decided in Milestone 6 under the configured tolerance (research R-09); ranking metrics are unaffected by a monotone map.

6. What this means for selection (Milestone 6)

PR-AUC and Recall@ K cannot discriminate between `hgb` (any set), `balanced_rf [primary]` and `logreg [primary]` on validation. The selection matrix must therefore lean on the other columns the spec requires: calibration quality, explainability, inference and maintenance cost (fit times 31 s, 72 s and 18 s respectively on 5.99 million rows), feature-set honesty (`hgb [strict_pretx]` is prediction-time safe and equally strong), and behaviour under the validation-to-test regime shift, which only the single-touch test evaluation can show. The headline feature set remains `primary` by project decision; the report will show `strict_pretx` beside it.

7. Caveats

- Six review periods is a small sample; per-period recall varies with the positive count alone.
- Perfect validation separability is a property of PaySim's generator and will not transfer to real banking data. Nothing here establishes real-world detection effectiveness.
- Threshold metrics use 0.5 pending the validation-chosen operating point.

Test discussion (task T059, single-touch evaluation, written 2026-09-05 after the tables above)

The test split (steps 553–743, 194,135 rows, 2,120 positives, prevalence 1.09%) was scored exactly once after the operating point was frozen; data/processed/test_access.json records the state and would record any re-evaluation with a reason. Every run was refitted on the full training split with its tuned parameters before scoring.

Validation-to-test shift. Prevalence rises from 0.83% to 1.09% and daily volume stays in the low-volume regime. Rankings are stable: hgb [primary] and hgb [posttx_ablation] keep PR-AUC 1.0000 (95% CI [1.0000, 1.0000]); balanced_rf on both sets holds at 0.9997; hgb [strict_pretx] moves from 1.0000 to 0.9995 [0.9986, 1.0000]; hgb [selected] from 0.9990 to 0.9996; logreg [primary] from 0.9987 to 0.9954 [0.9922, 0.9977]. The linear model on the strict set improves slightly (0.2776 → 0.2908) and the rule comparator improves from 0.1555 to 0.1856, both because higher prevalence makes precision easier. No candidate degrades materially, so the regime shift the split was designed to expose does not hurt tree models on this data.

Recall@200 is again a K ceiling. All strong models score 0.7568 (mean) / 0.7547 (pooled) with Precision@200 = 1.0000, because every test period holds 240–280 positives. The dummy (chronological) comparator rises to 0.2076 and random to 0.1012 only because day 31 contains 272 rows, all positives, where any order scores 0.735. See capacity_analysis.md.

Calibration on test. The reliability curves keep the class-weighting signature (observed rate below the diagonal in the middle bins), but the strong models place nearly all rows at scores near 0 or 1, so Brier stays tiny: hgb [primary] 0.0000 (3×10^{-6}), hgb [strict_pretx] 0.0000, balanced_rf [primary] 0.0003, logreg [primary] 0.0004. The validation-fitted isotonic calibrator, used only for the displayed probability, gives Brier 2.8×10^{-6} and ECE 6×10^{-6} on test.

Selected model at its operating point. hgb [primary] at the frozen raw-score threshold 0.9719: 2,115 true positives, 5 false negatives, 0 false positives across 192,015 normals (recall 0.9976, precision 1.0000). At the default 0.5 threshold it makes 0 errors of either kind on 194,135 rows.

Caveats carried forward. Eight review periods; one of them (day 31) is a partial day with 272 transactions. Perfect separation is a property of PaySim, not a transferable result. Bootstrap CIs resample rows and therefore understate uncertainty about future periods.

Model Selection Matrix

Method

Headline feature set: **primary** (project decision). Eligible rows are learners on the headline set; the verdict is decided from **validation** numbers only with the deterministic key val PR-AUC desc → val Recall@K desc → val Brier asc → explainability → fit s. Test numbers (single-touch evaluation with 95% bootstrap CIs) are reported beside the verdict and never used to choose it. Comparators and the dummy baseline appear in model_comparison.md, not here.

Matrix

candidate [set]	eligible	val PR-AUC	val Recall@200	val Precision@200	val Brier	val ECE	test PR-AUC	test PR-AUC 95% CI	test Recall@200	test pooled Recall@200 95% CI	explainability	fit s	investigator workload	verdict
hgb [primary]	yes	1.0000	0.8029	1.0000	0.0000	0.0000	1.0000	[1.0000, 1.0000]	0.7568	[0.7252, 0.7866]	medium: SHAP TreeExplainer exact; PDP/ICE valid on raw features	20.9200	val FP at 0.5 = 0; Precision@200 = 1.000	selected

candidate [set]	eligible	val PR-AUC	val Recall@200	val Precision@200	val Brier	val ECE	test PR-AUC	test PR-AUC 95% CI	test Recall@200	test pooled Recall@200 95% CI	explainability	fit s	investigator workload	verdict
balanced_rf [primary]	yes	1.0000	0.8029	1.0000	0.0004	0.0046	0.9997	[0.9992, 1.0000]	0.7568	[0.7252, 0.7866]	medium: SHAP TreeExplainer over 300 trees; slower to explain locally	135.7400	val FP at 0.5 = 5; Precision@200 = 1.000	eligible
logreg [primary]	yes	0.9987	0.8029	1.0000	0.0003	0.0065	0.9954	[0.9922, 0.9977]	0.7568	[0.7252, 0.7866]	high: linear coefficients on standardised features; SHAP linear explainer exact	18.6400	val FP at 0.5 = 5; Precision@200 = 1.000	eligible
hgb [posttx_ablation]		1.0000	0.8029	1.0000	0.0007	0.0024	1.0000	[1.0000, 1.0000]	0.7568	[0.7252, 0.7866]	medium: SHAP TreeExplainer exact; PDP/ICE valid on raw features	12.2500	val FP at 0.5 = 83; Precision@200 = 1.000	comparison only
hgb [strict_pretx]		1.0000	0.8029	1.0000	0.0000	0.0000	0.9995	[0.9986, 1.0000]	0.7568	[0.7252, 0.7866]	medium: SHAP TreeExplainer exact; PDP/ICE valid on raw features	23.6700	val FP at 0.5 = 0; Precision@200 = 1.000	comparison only
balanced_rf [strict_pretx]		1.0000	0.8029	1.0000	0.0008	0.0048	0.9997	[0.9994, 0.9999]	0.7568	[0.7252, 0.7866]	medium: SHAP TreeExplainer over 300 trees; slower to explain locally	152.6800	val FP at 0.5 = 116; Precision@200 = 1.000	comparison only
hgb [selected]		0.9990	0.8029	1.0000	0.0003	0.0006	0.9996	[0.9992, 0.9999]	0.7568	[0.7252, 0.7866]	medium: SHAP TreeExplainer exact; PDP/ICE valid on raw features	12.8600	val FP at 0.5 = 51; Precision@200 = 1.000	comparison only
hgb [pca_variant]		0.9069	0.7526	0.9400	0.0070	0.0113	0.9025	[0.8931, 0.9118]	0.7313	[0.7035, 0.7508]	medium: SHAP TreeExplainer exact; PDP/ICE valid on raw features	14.4800	val FP at 0.5 = 1,528; Precision@200 = 0.940	comparison only

candidate [set]	eligible	val PR-AUC	val Recall@200	val Precision@200	val Brier	val ECE	test PR-AUC	test PR-AUC 95% CI	test Recall@200	test pooled Recall@200 95% CI	explainability	fit s	investigator workload	verdict
logreg [strict_pretx]		0.2776	0.3117	0.3942	0.0589	0.0893	0.2908	[0.2729, 0.3072]	0.3539	[0.3426, 0.3743]	high: linear coefficients on standardised features; SHAP linear explainer exact	19.8300	val FP at 0.5 = 14,834; Precision@200 = 0.394	comparison only

Verdict reasoning (task T059, written 2026-09-05 after reviewing the matrix above)

Selected: hgb [primary] (histogram gradient boosting, tuned, on the headline primary set). The verdict was fixed on validation before the test split was unlocked and did not change afterwards.

Reading the matrix column by column:

- **PR-AUC (primary metric).** Three eligible candidates tie or nearly tie on validation: hgb [primary] 1.0000, balanced_rf [primary] 1.0000, logreg [primary] 0.9987. The deterministic key therefore moves to the next columns for the first two and drops the linear model on the primary metric alone. On test the order holds: 1.0000 [1.0000, 1.0000], 0.9997 [0.9992, 1.0000], 0.9954 [0.9922, 0.9977].
- **Recall@200 and Precision@200.** Identical for every eligible candidate on both splits (0.8029 on validation, 0.7568 on test, Precision@200 = 1.0000). Both are ceilings set by K (see capacity_analysis.md); this column cannot separate the strong candidates.
- **Calibration.** hgb [primary] has the lowest validation Brier and ECE (both rounding to 0.0000) versus 0.0004 / 0.0046 for balanced_rf [primary] and 0.0003 / 0.0065 for logreg [primary]. This is the column that decides between the two tied tree models.
- **Explainability.** The matrix ranks logistic regression higher (exact linear coefficients). Gradient boosting is rated medium: SHAP TreeExplainer is exact and PDP/ICE are valid on raw features, so investigator-facing explanations remain feasible (Milestone 7). Explainability was not reached by the key because calibration already separated the tied models; had it been reached it would have favoured the linear model over the forest.
- **Inference and maintenance cost.** Full-train fit time 20.9 s for hgb, 18.6 s for logreg, 135.7 s for balanced_rf (506 tuned trees). The forest is the most expensive to retrain and to explain locally.
- **Investigator workload.** At the 0.5 threshold on validation hgb [primary] produces 0 false positives; the forest and the linear model produce 5 each. All three fill the top 200 with positives.

Why not the strict pre-transaction set. hgb [strict_pretx] (comparison only) equals the selected model on validation (PR-AUC 1.0000, Brier 0.0000) and is within noise on test (0.9995 [0.9986, 1.0000]). It is prediction-time safe, which primary is not. The headline set is primary by project decision (batch triage framing, research R-06); the report presents the strict run beside the selected one, and a real-time deployment would be better served by it.

Feature-set honesty. hgb [posttx_ablation], built only from type plus the five post-transaction fields, also reaches 1.0000 on both splits. Together with strict_pretx at 0.9995 this shows that both the artifact features and the behavioural features are individually sufficient to reproduce PaySim's label; the selected model's performance is not evidence of transferable AML skill.

Process notes. The operating point was frozen twice before any test access: the first freeze used a threshold of 1.0 produced by isotonic calibration collapsing validation scores to 0/1; the rule was corrected to threshold and rank on raw scores and the split was re-frozen (recorded under `refreezes` in `data/processed/test_access.json`). Two evaluation attempts were killed by the operating system for memory before any run finished the protocol; the state never reached evaluated until the successful single run, and no test result was seen before the operating point was final.

Capacity Analysis

Scope

Selected run `hgb__primary`; review period = 24 steps (one simulated day); capacity $K = 200$ with sensitivity grid [50, 100, 200, 300, 500]. Recall@K = share of a period's positives inside the top-K; Precision@K = share of the top-K that are positives. Business figures are **illustrative counts** on synthetic data.

val: Recall@K and Precision@K across the capacity grid

K	Recall@K mean/period	Recall@K pooled	Precision@K mean/period	Precision@K pooled
50	0.2007	0.1995	1.0000	1.0000
100	0.4015	0.3989	1.0000	1.0000
200	0.8029	0.7979	1.0000	1.0000
300	1.0000	1.0000	0.8356	0.8356
500	1.0000	1.0000	0.5013	0.5013

val: per review period at $K = 200$

day	steps	transactions	positives	reviewed (k_eff)	positives caught	positives missed (FN)	reviews spent on normals (FP)	Recall@K	Precision@K
18	409–432	20,999	268	200	200	68	0	0.7463	1.0000
19	433–456	11,300	256	200	200	56	0	0.7812	1.0000
20	457–480	19,727	236	200	200	36	0	0.8475	1.0000
21	481–504	24,593	272	200	200	72	0	0.7353	1.0000
22	505–528	53,437	256	200	200	56	0	0.7812	1.0000
23	529–552	51,012	216	200	200	16	0	0.9259	1.0000

val: illustrative KPI — positives surfaced per review period at $K = 200$

ranking	illustrative positives surfaced per day	improvement factor vs selected	Recall@200
hgb [primary] (selected)	200.0000	1.0000	0.8029
rule comparator (flag, then amount)	50.0000	4.0000	0.1965
random ranking	2.7000	75.0000	0.0104
dummy (chronological order)	16.7000	12.0000	0.0650

Illustrative counts on synthetic data; not a real-world estimate and never expressed in currency.

test: Recall@K and Precision@K across the capacity grid

K	Recall@K mean/period	Recall@K pooled	Precision@K mean/period	Precision@K pooled
50	0.1892	0.1887	1.0000	1.0000
100	0.3784	0.3774	1.0000	1.0000
200	0.7568	0.7547	1.0000	1.0000

K	Recall@K mean/period	Recall@K pooled	Precision@K mean/period	Precision@K pooled
300	1.0000	1.0000	0.8950	0.8938
500	1.0000	1.0000	0.5870	0.5620

test: per review period at K = 200

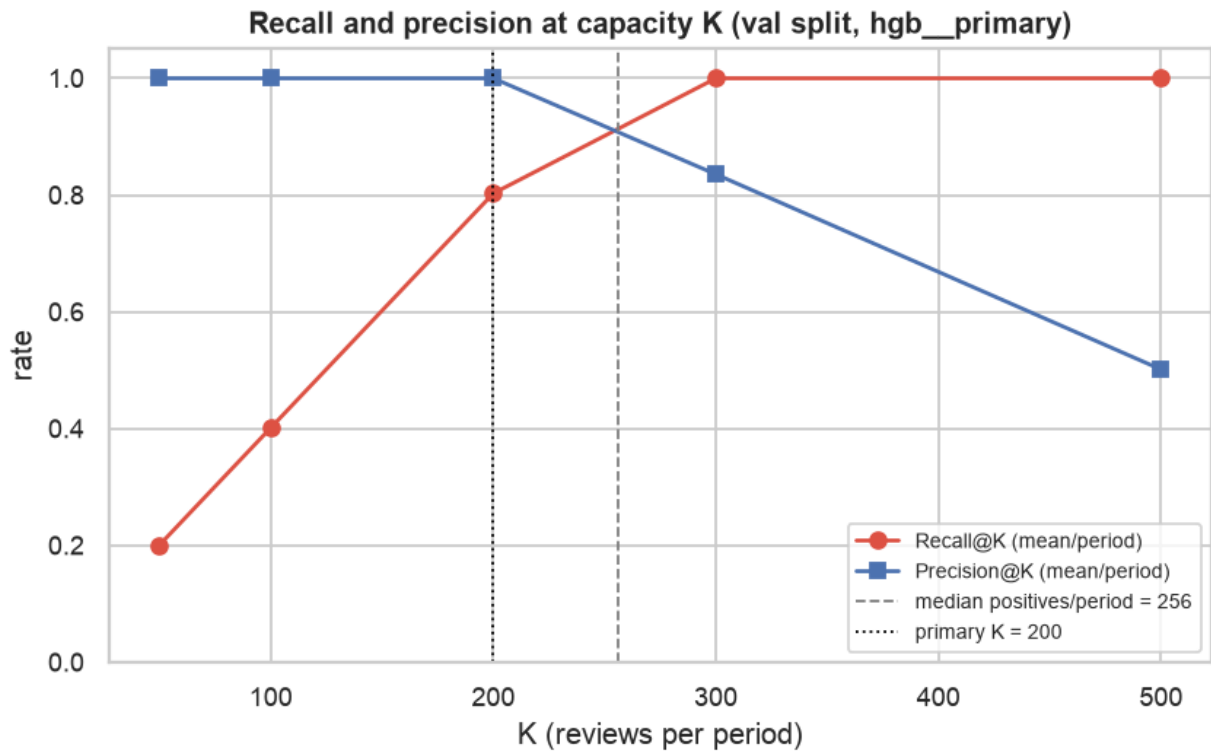
day	steps	transactions	positives	reviewed (k_eff)	positives caught	positives missed (FN)	reviews spent on normals (FP)	Recall@K	Precision@K
24	553–576	32,709	280	200	200	80	0	0.7143	1.0000
25	577–600	57,853	240	200	200	40	0	0.8333	1.0000
26	601–624	13,885	272	200	200	72	0	0.7353	1.0000
27	625–648	8,578	280	200	200	80	0	0.7143	1.0000
28	649–672	14,661	248	200	200	48	0	0.8065	1.0000
29	673–696	54,890	260	200	200	60	0	0.7692	1.0000
30	697–720	11,287	268	200	200	68	0	0.7463	1.0000
31	721–743	272	272	200	200	72	0	0.7353	1.0000

test: illustrative KPI — positives surfaced per review period at K = 200

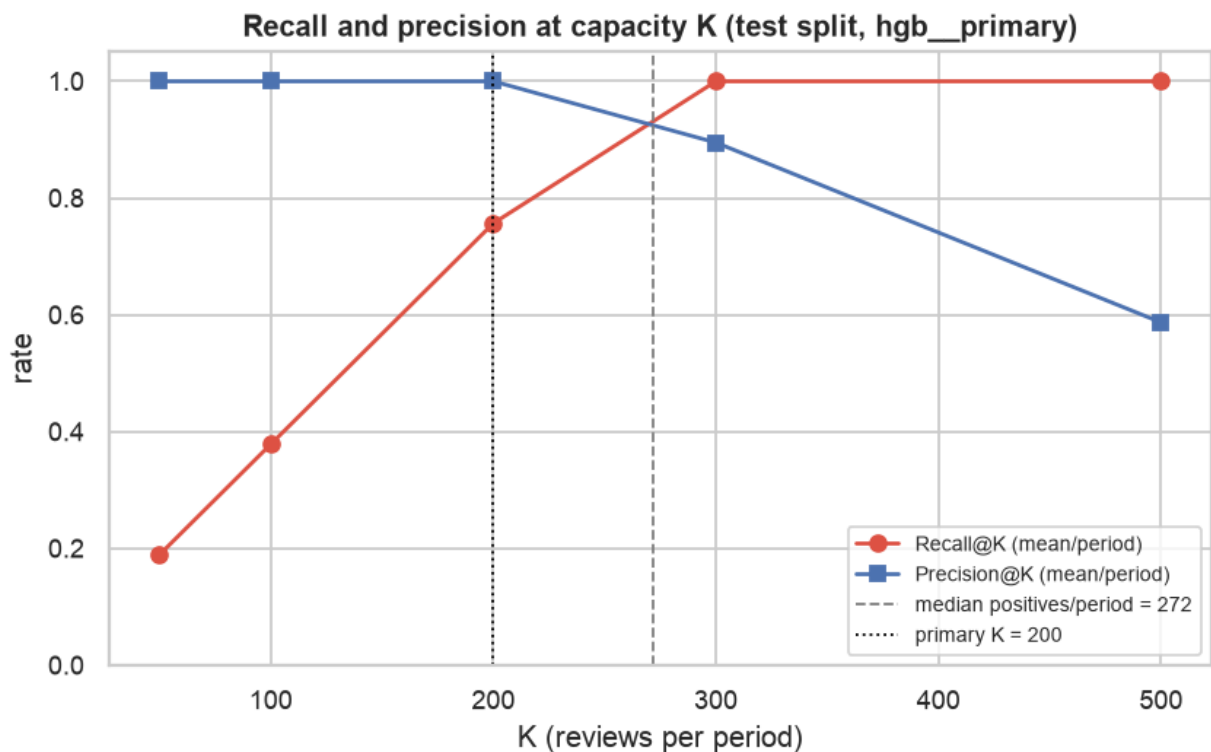
ranking	illustrative positives surfaced per day	improvement factor vs selected	Recall@200
hgb [primary] (selected)	200.0000	1.0000	0.7568
rule comparator (flag, then amount)	83.2000	2.4000	0.3101
random ranking	27.5000	7.3000	0.1012
dummy (chronological order)	56.5000	3.5000	0.2076

Illustrative counts on synthetic data; not a real-world estimate and never expressed in currency.

Figures



Recall rises with K until K exceeds the positives in a period; precision falls once K passes that point.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.



Recall rises with K until K exceeds the positives in a period; precision falls once K passes that point.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.

Reading the capacity tables (task T059, written 2026-09-05 after reviewing the tables above)

Recall@K is a ceiling set by K, on test as on validation. Every test review period (days 24–31) holds between 240 and 280 positives, all above $K = 200$. With $\text{Precision}@200 = 1.0000$ in all eight periods, the selected model catches exactly 200 positives per period and misses the rest: $\text{Recall}@200 = \text{mean}(200 / \text{positives}) = 0.7568$ (pooled 0.7547, 95% bootstrap CI [0.7252, 0.7866]). At $K = 300$ every positive is caught (Recall 1.0000) and $\text{Precision}@300$ falls to 0.8950 because the remaining reviews land on normals; at $K = 500$ precision is 0.5870. The capacity curve crosses near the median of 272 positives per period. $\text{Recall}@50$ and $\text{Recall}@100$ are exactly 50 and 100 divided by the positives per period (0.1892 and 0.3784).

Validation to test. Recall@200 drops from 0.8029 on validation to 0.7568 on test only because test periods contain more positives (240–280 versus 216–272); the model's ranking is perfect in both. Prevalence rises from 0.83% to 1.09%; PR-AUC stays at 1.0000.

False positives and false negatives at K = 200. In every period the 200 reviewed transactions are all positives (0 reviews spent on normals) and 40–80 positives per day are not reviewed (FN), totalling 520 unreviewed positives over the eight test days. With this model, the trade-off is not between reviewing normals and missing positives; it is purely a capacity question. Raising K from 200 to 300 would clear the backlog at the cost of about 30 reviews per day spent on normals.

Threshold-based operating point. At the frozen raw-score threshold of 0.9719 the selected model flags 2,115 of 2,120 test positives with 0 false positives across 192,015 normals (recall 0.9976, FPR 0.0000). The 5 positives below threshold are still ranked above every normal, so the ranked queue loses nothing; the threshold only governs the medium/low priority labels.

Illustrative KPI (synthetic counts, never currency). At K = 200 the selected model surfaces 200 positives per day. The rule comparator (flag, then amount) surfaces 83.2 (2.4× fewer), chronological order 56.5 (3.5× fewer) and random ranking 27.5 (7.3× fewer). Chronological order looks better than random only because day 31 contains 272 transactions that are all positives, so any ordering scores 0.735 there.

Success criteria. SC-001: Recall@200 of 0.7568 exceeds random ranking (0.1012) and the dummy baseline (0.2076), and PR-AUC 1.0000 exceeds the no-skill value of 0.0109. SC-002: the rule comparator reaches 0.3101; the selected model exceeds it. Both criteria are met on the single-touch test evaluation.

What these numbers do not mean. PaySim positives are generated by a rule the features reproduce exactly. Perfect precision at capacity says the generator is easy to invert, not that a real AML queue would look like this. Real transaction data would show far lower precision, positives that do not share one signature, and drifting prevalence; the capacity analysis method carries over, the figures do not.

5. Explainability

Explainability

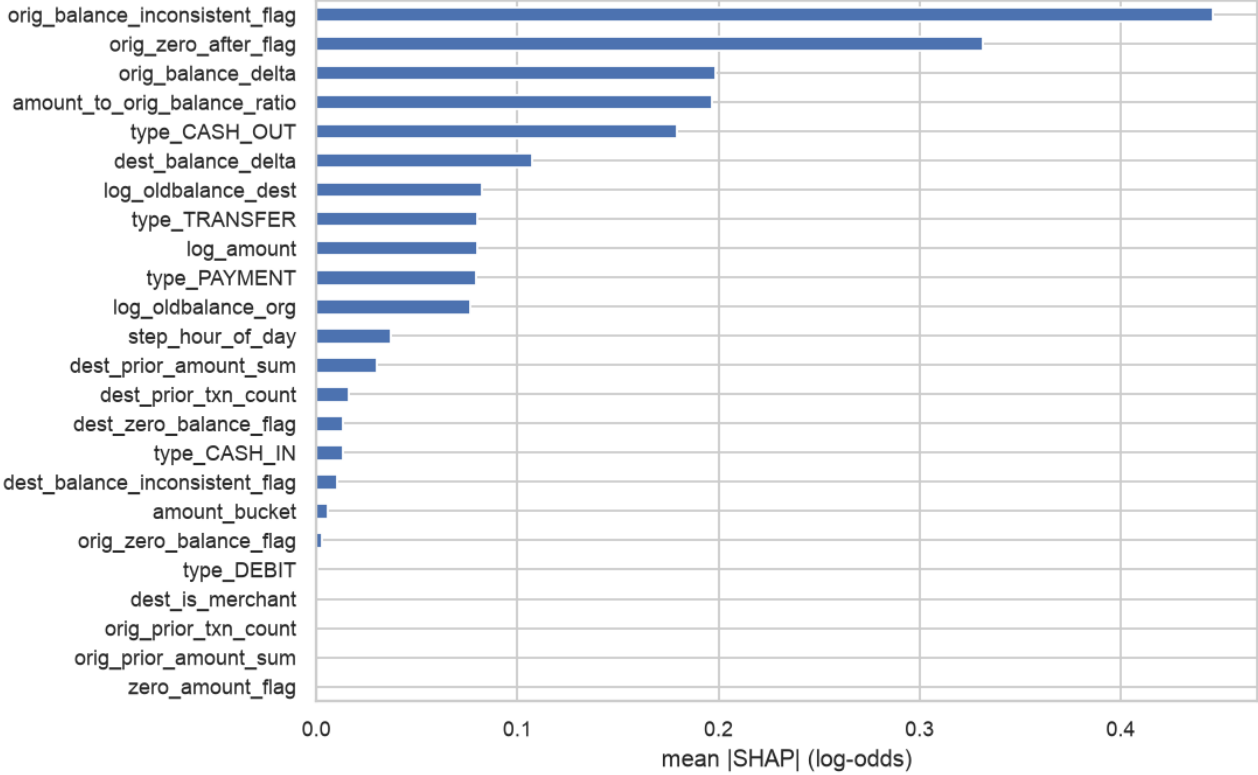
Scope

Released bundle 20260904T225142-0dc8f82-hgb (hgb on primary). Explainer: TreeExplainer (exact), contributions in log-odds. Background: 1,000 seeded training rows; global sample: 2,000 seeded test rows; local examples: the top-ranked transactions of the first test review period. Explanations describe the model, not the transactions' true nature.

Global

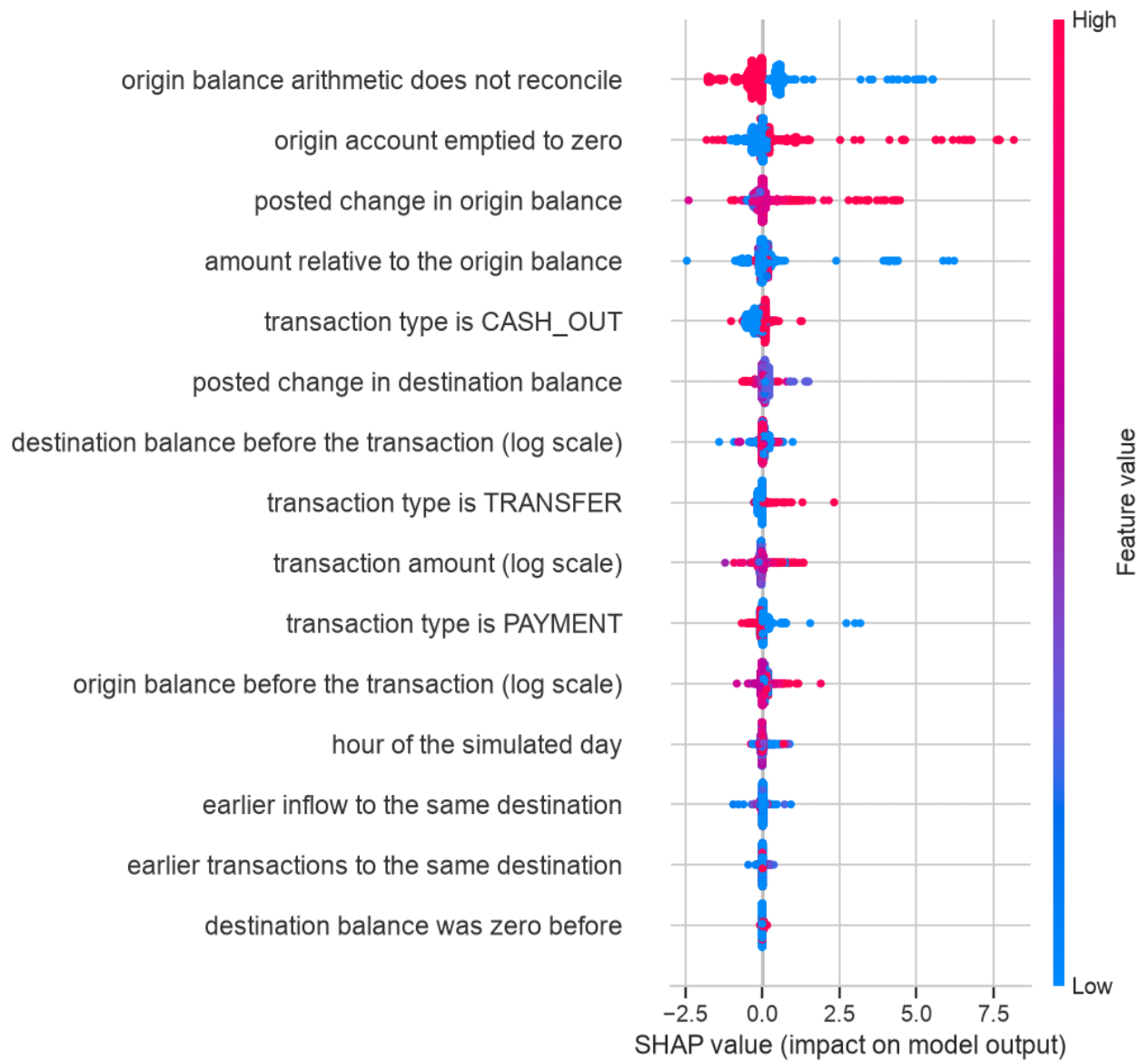
feature	mean SHAP (log-odds)	registry rationale
orig_balance_inconsistent_flag	0.4458	Direction-aware arithmetic gap on the origin side is inconsistent for most CASH_OUT/TRANSFER rows (DQ-05); a simulator behaviour that correlates with the label.
orig_zero_after_flag	0.3313	Account emptied to exactly zero after the transaction (DQ-05); a strong mule pattern in the simulator.
orig_balance_delta	0.1984	Posted change in origin balance; in batch triage the posted state is available and its mismatch with amount is informative (DQ-05).
amount_to_orig_balance_ratio	0.1966	Emptying an account (ratio near 1) is a classic mule-account pattern; the +1 guard handles zero balances.
type_CASH_OUT	0.1795	Simulated fraud occurs only in TRANSFER and CASH_OUT (DQ-09); type is the first-order signal.
dest_balance_delta	0.1075	Posted change in destination balance (DQ-06).
log_oldbalance_dest	0.0826	Destination balance before the transaction; merchants carry zero (DQ-06), customers vary widely.
type_TRANSFER	0.0802	Simulated fraud occurs only in TRANSFER and CASH_OUT (DQ-09); type is the first-order signal.
log_amount	0.0798	Amounts are heavy-tailed (DQ-07); log scale stabilises linear models and keeps tree splits interpretable.
type_PAYMENT	0.0796	Simulated fraud occurs only in TRANSFER and CASH_OUT (DQ-09); type is the first-order signal.
log_oldbalance_org	0.0768	Origin balance before the transaction is heavy-tailed with a mass at zero (DQ-07); log scale plus a zero flag captures both.
step_hour_of_day	0.0372	Hourly position within the simulated day may carry volume seasonality; cyclic and in-range for every split.
dest_prior_amount_sum	0.0304	Cumulative earlier inflow to the same destination (DQ-11).
dest_prior_txn_count	0.0161	Destinations repeat (16.9% appear more than once, max 113; DQ-11); accumulation into one destination is a laundering-style pattern.
dest_zero_balance_flag	0.0134	Zero destination balance before the transaction marks merchants and fresh accounts (DQ-06).

Global feature importance, 20260904T225142-0dc8f82-hgb (seeded test sample n=2,000)



Features that move the risk score most across the sampled test transactions (mean |SHAP| in log-odds): origin balance arithmetic does not reconcile (0.446), origin account emptied to zero (0.331), posted change in origin balance (0.198), amount relative to the origin balance (0.197), transaction type is CASH_OUT (0.180). Educational decision-support prototype trained on synthetic PaySim data. Outputs are risk scores and review priorities that help human investigators decide what to review first. This system makes no fraud or AML determination and performs no automatic blocking, account closure, customer risk rating, or regulatory reporting. Results on synthetic data do not establish real-world detection effectiveness, fairness, or regulatory suitability.
Synthetic PaySim data; educational decision-support prototype; no AML determination.

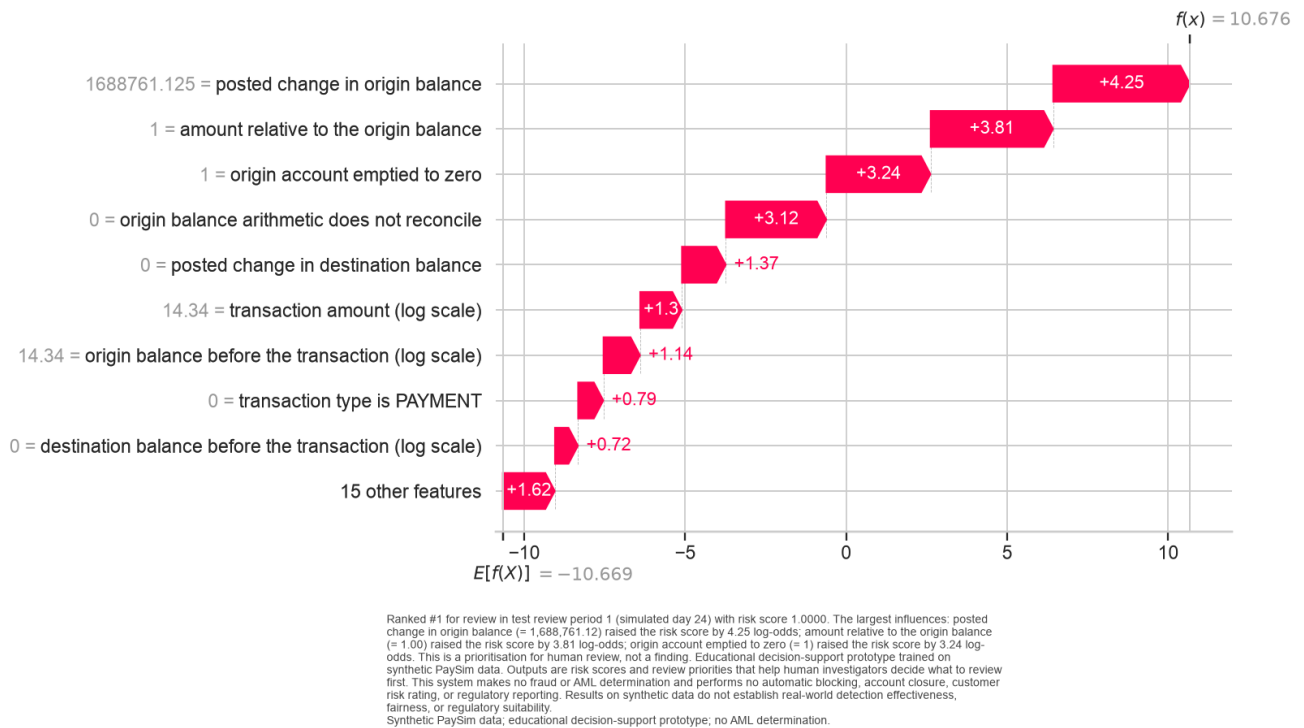
SHAP summary (beeswarm), 20260904T225142-0dc8f82-hgb



Each point is one sampled test transaction; colour is the feature value; x is the contribution in log-odds.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.

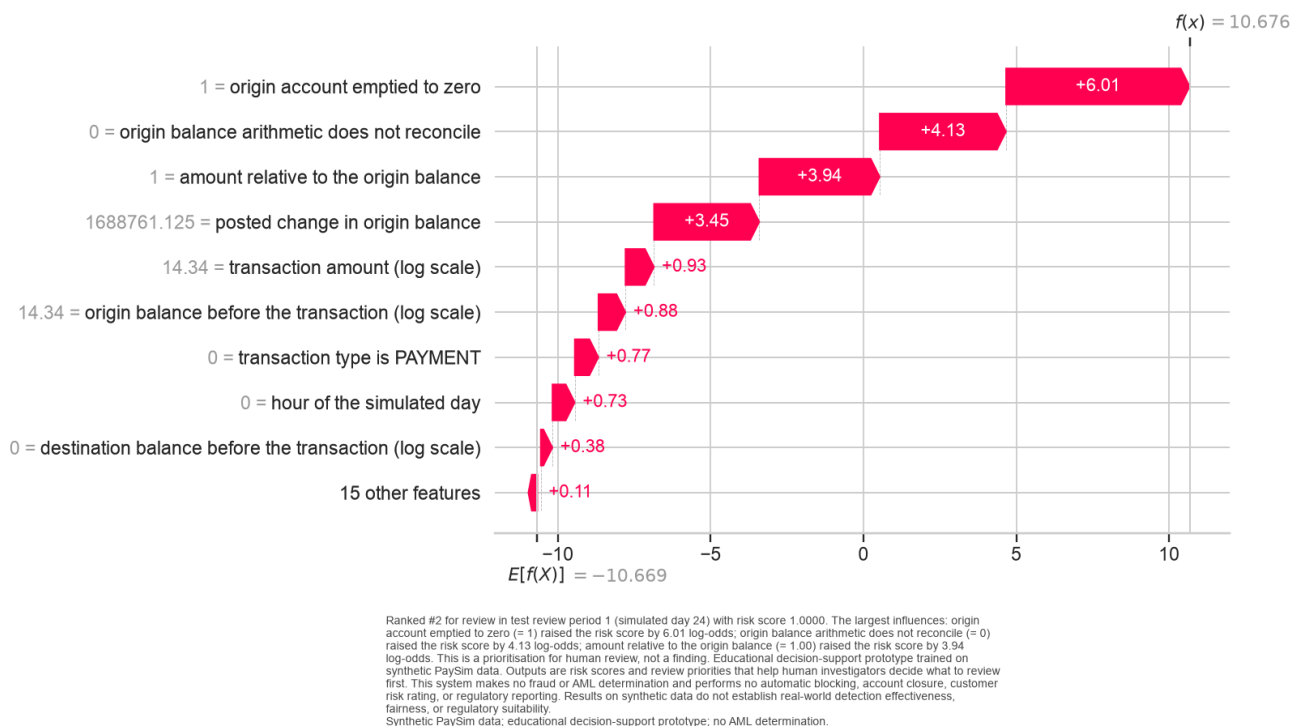
Local Examples

Rank 1 (row 6,168,485, step 553, TRANSFER, score 1.0000)



Ranked #1 for review in test review period 1 (simulated day 24) with risk score 1.0000. The largest influences: posted change in origin balance (= 1,688,761.12) raised the risk score by 4.25 log-odds; amount relative to the origin balance (= 1.00) raised the risk score by 3.81 log-odds; origin account emptied to zero (= 1) raised the risk score by 3.24 log-odds. This is a prioritisation for human review, not a finding. Educational decision-support prototype trained on synthetic PaySim data. Outputs are risk scores and review priorities that help human investigators decide what to review first. This system makes no fraud or AML determination and performs no automatic blocking, account closure, customer risk rating, or regulatory reporting. Results on synthetic data do not establish real-world detection effectiveness, fairness, or regulatory suitability.

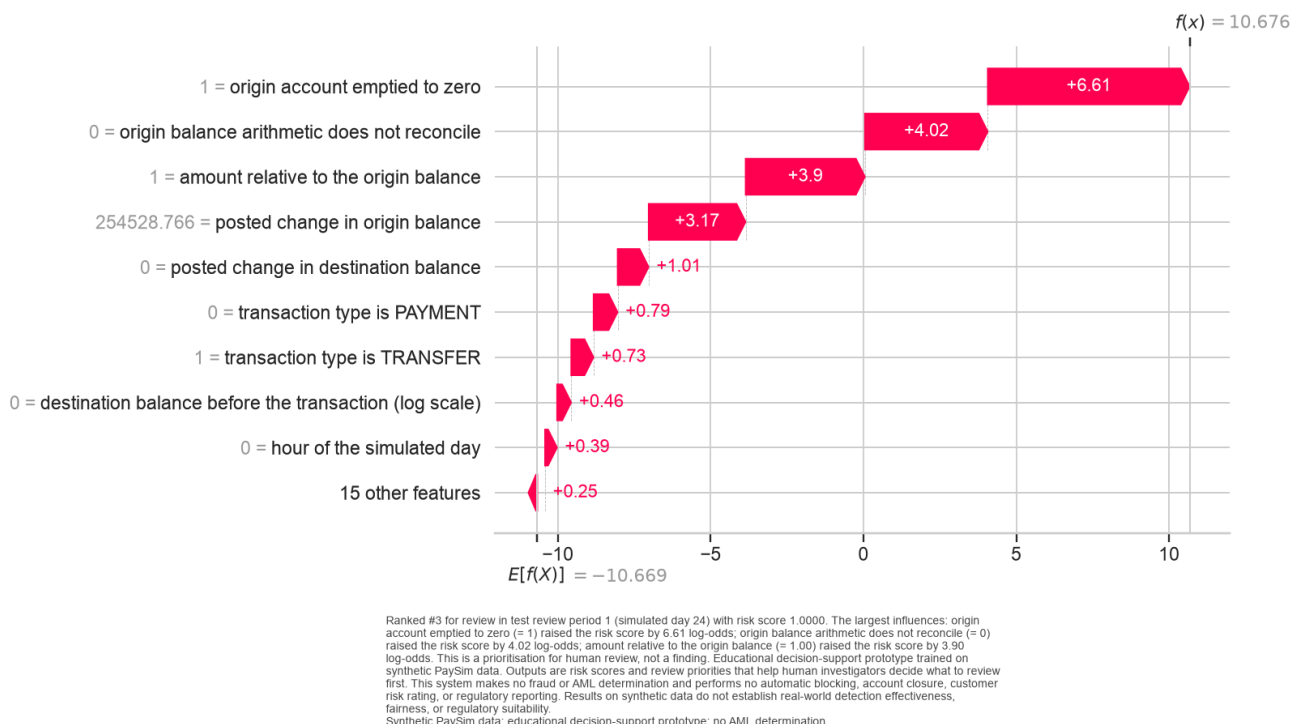
Rank 2 (row 6,168,486, step 553, CASH_OUT, score 1.0000)



Ranked #2 for review in test review period 1 (simulated day 24) with risk score 1.0000. The largest influences: origin account emptied to zero (= 1) raised the risk score by 6.01 log-odds; origin balance arithmetic does not reconcile (= 0) raised the risk score by 4.13 log-odds; amount relative to the origin balance (= 1.00) raised the risk score by 3.94 log-odds. This is a prioritisation for human review, not a finding. Educational decision-support prototype trained on

synthetic PaySim data. Outputs are risk scores and review priorities that help human investigators decide what to review first. This system makes no fraud or AML determination and performs no automatic blocking, account closure, customer risk rating, or regulatory reporting. Results on synthetic data do not establish real-world detection effectiveness, fairness, or regulatory suitability.

Rank 3 (row 6,168,487, step 553, TRANSFER, score 1.0000)

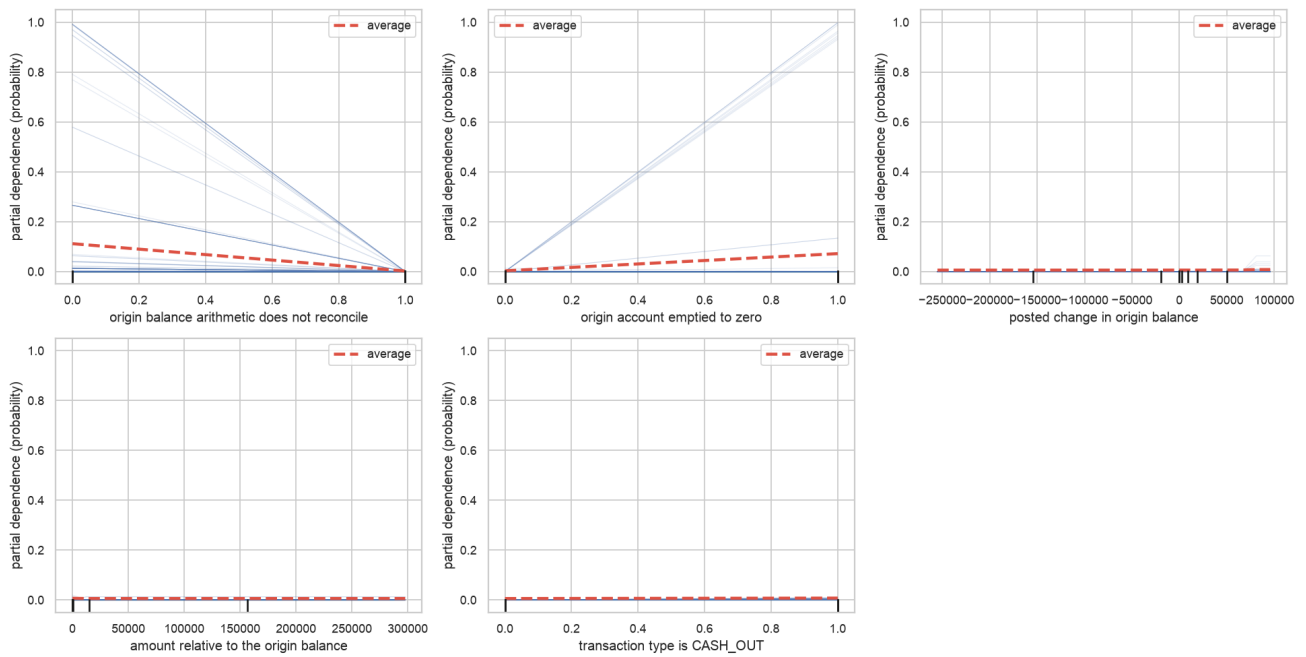


Ranked #3 for review in test review period 1 (simulated day 24) with risk score 1.0000. The largest influences: origin account emptied to zero (= 1) raised the risk score by 6.61 log-odds; origin balance arithmetic does not reconcile (= 0) raised the risk score by 4.02 log-odds; amount relative to the origin balance (= 1.00) raised the risk score by 3.90 log-odds. This is a prioritisation for human review, not a finding. Educational decision-support prototype trained on synthetic PaySim data. Outputs are risk scores and review priorities that help human investigators decide what to review first. This system makes no fraud or AML determination and performs no automatic blocking, account closure, customer risk rating, or regulatory reporting. Results on synthetic data do not establish real-world detection effectiveness, fairness, or regulatory suitability.

PDP/ICE Validity

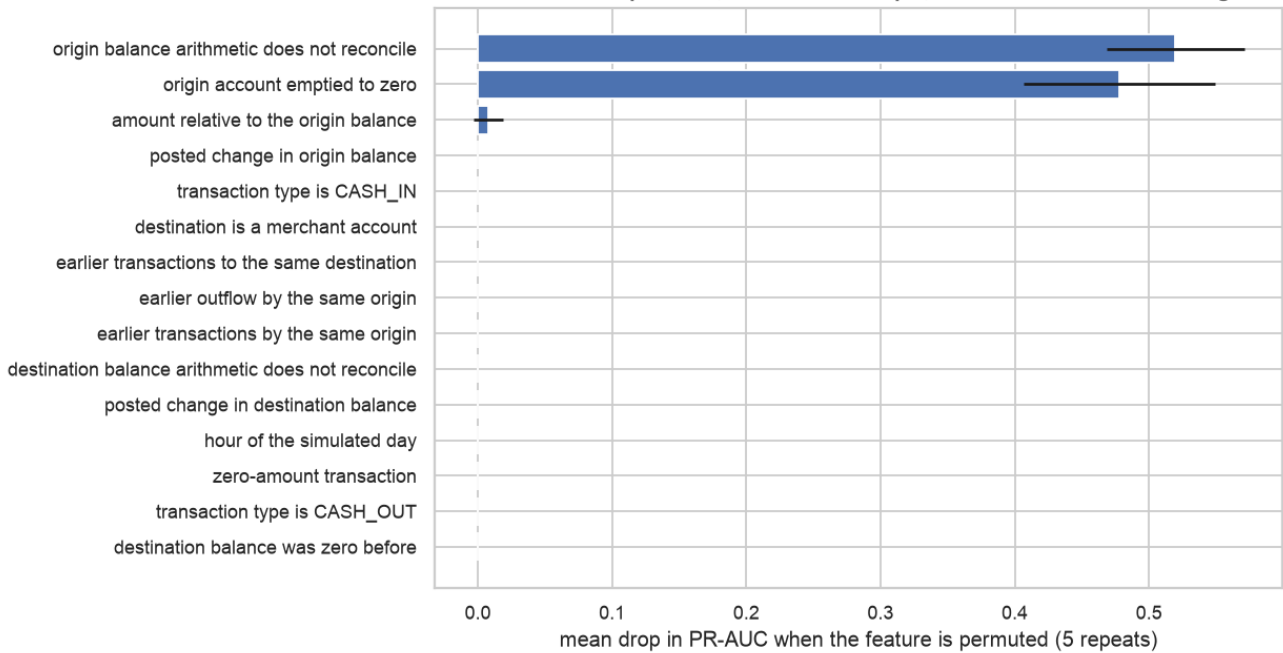
feature	status	reason	alternative
orig_balance_inconsistent_flag	produced	binary flag: the curve has two points	
orig_zero_after_flag	produced	binary flag: the curve has two points	
orig_balance_delta	produced	max Spearman ρ with other top features = 0.58	
amount_to_orig_balance_ratio	produced	max Spearman ρ with other top features = 0.47	
type_CASH_OUT	produced	binary flag: the curve has two points	

PDP (red) and ICE (thin) on a seeded test sample, 20260904T225142-0dc8f82-hgb



Average (PDP) and individual (ICE) response of the predicted probability to each feature; features strongly correlated with another top feature are omitted (see validity table).
Synthetic PaySim data; educational decision-support prototype; no AML determination.

Permutation importance, seeded test sample, 20260904T225142-0dc8f82-hgb



Model-agnostic alternative to PDP for correlated features; a near-zero drop means the model can recover the signal from correlated features.
Synthetic PaySim data; educational decision-support prototype; no AML determination.

Permutation importance (alternative / cross-check)

feature	mean drop in PR-AUC	std
orig_balance_inconsistent_flag	0.5195	0.0513
orig_zero_after_flag	0.4781	0.0711
amount_to_orig_balance_ratio	0.0078	0.0111
orig_balance_delta	0.0000	0.0000
type_CASH_IN	0.0000	0.0000
dest_is_merchant	0.0000	0.0000

feature	mean drop in PR-AUC	std
dest_prior_txn_count	0.0000	0.0000
orig_prior_amount_sum	0.0000	0.0000
orig_prior_txn_count	0.0000	0.0000
dest_balance_inconsistent_flag	0.0000	0.0000
dest_balance_delta	0.0000	0.0000
step_hour_of_day	0.0000	0.0000
zero_amount_flag	0.0000	0.0000
type_CASH_OUT	0.0000	0.0000
dest_zero_balance_flag	0.0000	0.0000

Consistency Notes (task T071, written 2026-09-05 after reviewing the figures and tables above)

Agreement with EDA, feature by feature (top five by mean |SHAP|).

- `orig_balance_inconsistent_flag` (0.446, rank 1). Beeswarm: the flag at 0 (blue) pushes the score up, at 1 (red) pulls it down; the PDP falls from about 0.11 to 0 as the flag goes 0 → 1. This matches `eda_08`: positives have exact origin bookkeeping (positive rate 0.27% when the arithmetic reconciles versus 0.0016% when it does not). It is the simulator artifact DQ-05, and the model's single most important input.
- `orig_zero_after_flag` (0.331, rank 2). Value 1 raises the score; some ICE lines go from 0 to 1. Matches `eda_08` (positive rate 0.32% versus 0.002%) and the diagonal in `eda_09`.
- `orig_balance_delta` (0.198, rank 3). Large posted changes raise the score (local example 1: a delta of 1,688,761 contributes +4.25 log-odds). Consistent with `eda_07`, where positives carry a long right tail. Its PDP is flat because the effect only appears jointly with the two flags above; the ICE spread shows that interaction.
- `amount_to_orig_balance_ratio` (0.197, rank 4). Ratios near 1 raise the score (all three local examples sit at exactly 1.00). Matches `eda_07` and `eda_09` (the "empty the account" diagonal). The PDP grid is dominated by extreme ratios from near-zero balances and shows nothing in the informative region near 1; this is a limitation of the grid, not evidence of no effect.
- `type_CASH_OUT` (0.180, rank 5). Consistent with `eda_01`: positives exist only in CASH_OUT and TRANSFER.

Surprises, not omitted.

- `dest_is_merchant`, `zero_amount_flag`, `type_DEBIT` and both origin aggregates have mean |SHAP| of 0.000. For `dest_is_merchant` this is redundancy with `type_PAYMENT`, not irrelevance. For `zero_amount_flag` it is scarcity: 4 training rows.
- Permutation importance shows the model depends on exactly two features. Permuting `orig_balance_inconsistent_flag` drops PR-AUC by 0.52 and `orig_zero_after_flag` by 0.48; every other feature permutes to a drop of 0.008 or less because the remaining signal is recoverable from correlated columns. The released model is, in effect, the rule "origin bookkeeping reconciles and the origin account was emptied".
- Three of the top four features are post-transaction (batch-only) fields. This is the artifact dominance anticipated in research R-06. The `strict_pretx` run reached the same PR-AUC without them, so the behavioural signal exists, but the released model prefers the bookkeeping shortcut.

The five test positives below the operating-point threshold. Three are zero-amount CASH_OUT rows with zero balances (raw scores 0.80–0.92); two are TRANSFERS whose posted origin balance did not change (10,399,045 → 10,399,045 and 5,674,548 → 5,674,548; scores 0.52 and 0.97). All five are generator edge cases rather than behavioural misses, and all five still rank above every normal transaction in their period.

Plain-language summary for a business audience

The model ranks a transaction near the top when the sending account is emptied by the transaction and the posted balances reconcile exactly, especially for transfers and cash-outs of large amounts relative to the balance. On this synthetic data that pattern identifies almost every simulated positive. A real bank's data would not hand the model such a clean bookkeeping signature, so the explanation method transfers; the specific features that dominate here do not. Investigators see, for each queued transaction, the three factors that moved its score most and can override the ranking.

6. Bias & Fairness Analysis

Bias & Fairness Analysis

Sensitive-Attribute Availability Record

Checked on 2026-09-05 against the actual raw columns of PS_20174392719_1491204439457_log.csv: step, type, amount, nameOrig, oldbalanceOrig, newbalanceOrig, nameDest, oldbalanceDest, newbalanceDest, isFraud, isFlaggedFraud. Proxy scan terms: age, birth, citizen, country, dob, education, ethnicity, gender, income, marital, nationality, occupation, postcode, race, region, religion, sex, socioeconomic, wealth, zip; matching columns: none.

attribute	valid label present	evidence
age	no	no column among ['step', 'type', 'amount', 'nameOrig', 'oldbalanceOrig', 'newbalanceOrig', 'nameDest', 'oldbalanceDest', 'newbalanceDest', 'isFraud', 'isFlaggedFraud'] contains any of ['age', 'birth', 'dob']
ethnicity	no	no column among ['step', 'type', 'amount', 'nameOrig', 'oldbalanceOrig', 'newbalanceOrig', 'nameDest', 'oldbalanceDest', 'newbalanceDest', 'isFraud', 'isFlaggedFraud'] contains any of ['ethnicity', 'race']
gender	no	no column among ['step', 'type', 'amount', 'nameOrig', 'oldbalanceOrig', 'newbalanceOrig', 'nameDest', 'oldbalanceDest', 'newbalanceDest', 'isFraud', 'isFlaggedFraud'] contains any of ['gender', 'sex']
nationality	no	no column among ['step', 'type', 'amount', 'nameOrig', 'oldbalanceOrig', 'newbalanceOrig', 'nameDest', 'oldbalanceDest', 'newbalanceDest', 'isFraud', 'isFlaggedFraud'] contains any of ['nationality', 'country', 'citizen']
socioeconomic_status	no	no column among ['step', 'type', 'amount', 'nameOrig', 'oldbalanceOrig', 'newbalanceOrig', 'nameDest', 'oldbalanceDest', 'newbalanceDest', 'isFraud', 'isFlaggedFraud'] contains any of ['income', 'socioeconomic', 'wealth', 'occupation', 'education', 'region', 'zip', 'postcode']

any_valid_label = false

Demographic Fairness

Demographic fairness metrics cannot be computed on this dataset because no valid sensitive-group labels exist. What follows is an operational error-slice analysis over non-protected partitions of the data; it is not a fairness measurement across protected groups and must not be described as one.

Operational Error-Slice Analysis

Label: **operational error-slice analysis**. Test split; K = 200; raw-score threshold 0.971931; amount and origin-balance band edges fitted on the training split ({'amount_band': [9952.751953125, 36656.2625, 122966.15625, 247127.65625], 'orig_balance_band': [13767.0, 51415.0, 301979.0]}). Recall@K within a slice is the share of that slice's positives that fall inside their review period's top-K.

By type

slice	rows	positives	prevalence	Recall@200	Precision@200	FNR at threshold	FPR at threshold	Brier (calibrated)
CASH_IN	44,564	0	0.0000				0.0000	0.0000
CASH_OUT	63,155	1,060	0.0168	0.7689	1.0000	0.0028	0.0000	0.0000
DEBIT	1,510	0	0.0000				0.0000	0.0000
PAYMENT	66,334	0	0.0000				0.0000	0.0000

slice	rows	positives	prevalence	Recall@200	Precision@200	FNR at threshold	FPR at threshold	Brier (calibrated)
TRANSFER	18,572	1,060	0.0571	0.7406	1.0000	0.0019	0.0000	0.0000

By amount_band

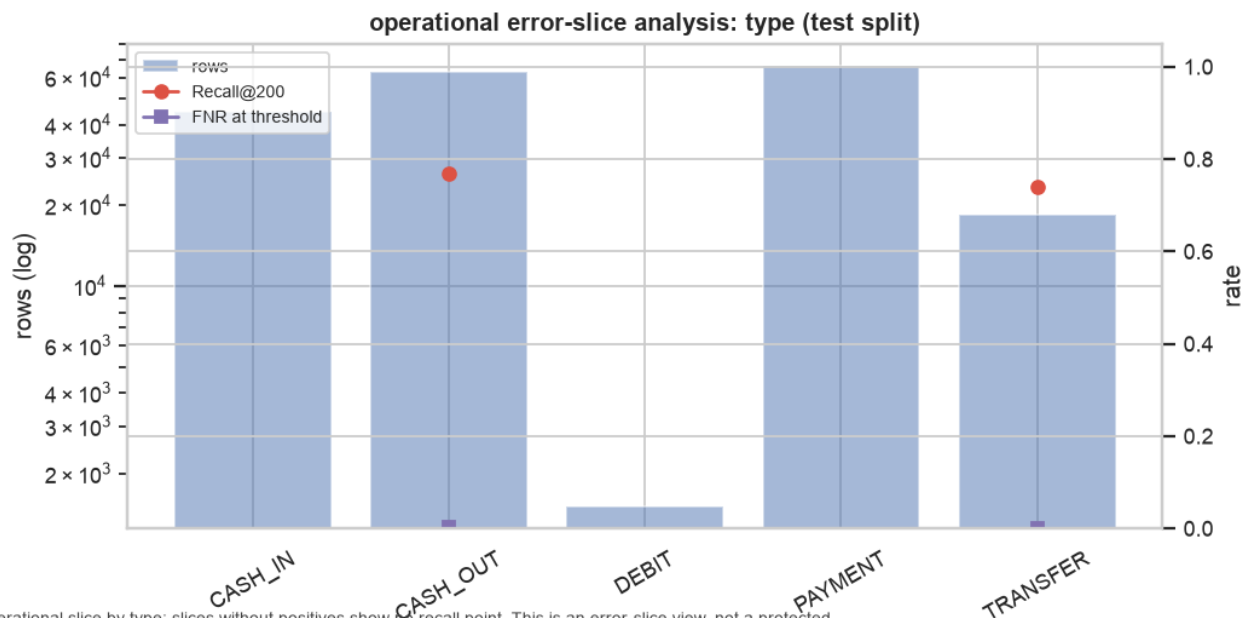
slice	rows	positives	prevalence	Recall@200	Precision@200	FNR at threshold	FPR at threshold	Brier (calibrated)
high	38,717	260	0.0067	0.9615	1.0000	0.0000	0.0000	0.0000
low	37,166	106	0.0029	0.0000		0.0000	0.0000	0.0000
mid	37,413	342	0.0091	0.5409	1.0000	0.0000	0.0000	0.0000
very high	37,382	1,338	0.0358	0.8707	1.0000	0.0015	0.0000	0.0000
very low	43,457	74	0.0017	0.0000		0.0405	0.0000	0.0000

By orig_balance_band

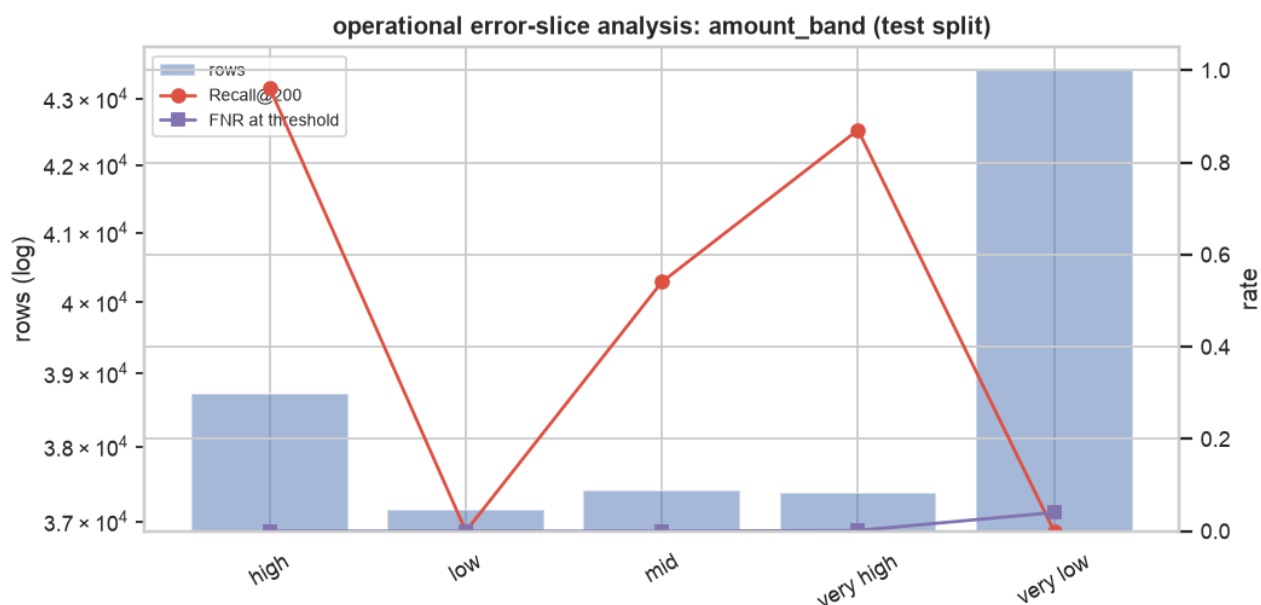
slice	rows	positives	prevalence	Recall@200	Precision@200	FNR at threshold	FPR at threshold	Brier (calibrated)
Q1	36,916	72	0.0020	0.0000		0.0000	0.0000	0.0000
Q2	36,494	164	0.0045	0.0488	1.0000	0.0000	0.0000	0.0000
Q3	34,673	634	0.0183	0.8186	1.0000	0.0000	0.0000	0.0000
Q4	32,937	1,240	0.0376	0.8653	1.0000	0.0016	0.0000	0.0000
zero	53,115	10	0.0002	0.0000		0.3000	0.0000	0.0000

By step_band

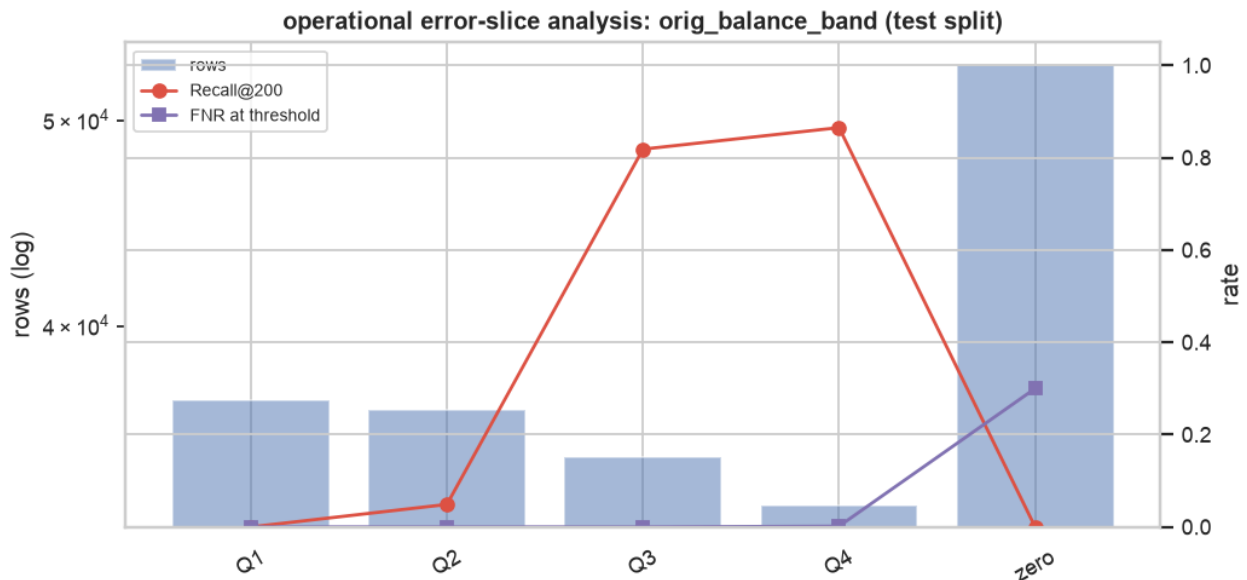
slice	rows	positives	prevalence	Recall@200	Precision@200	FNR at threshold	FPR at threshold	Brier (calibrated)
day 24	32,709	280	0.0086	0.7143	1.0000	0.0000	0.0000	0.0000
day 25	57,853	240	0.0041	0.8333	1.0000	0.0000	0.0000	0.0000
day 26	13,885	272	0.0196	0.7353	1.0000	0.0000	0.0000	0.0000
day 27	8,578	280	0.0326	0.7143	1.0000	0.0107	0.0000	0.0001
day 28	14,661	248	0.0169	0.8065	1.0000	0.0040	0.0000	0.0000
day 29	54,890	260	0.0047	0.7692	1.0000	0.0000	0.0000	0.0000
day 30	11,287	268	0.0237	0.7463	1.0000	0.0000	0.0000	0.0000
day 31	272	272	1.0000	0.7353	1.0000	0.0037		0.0000



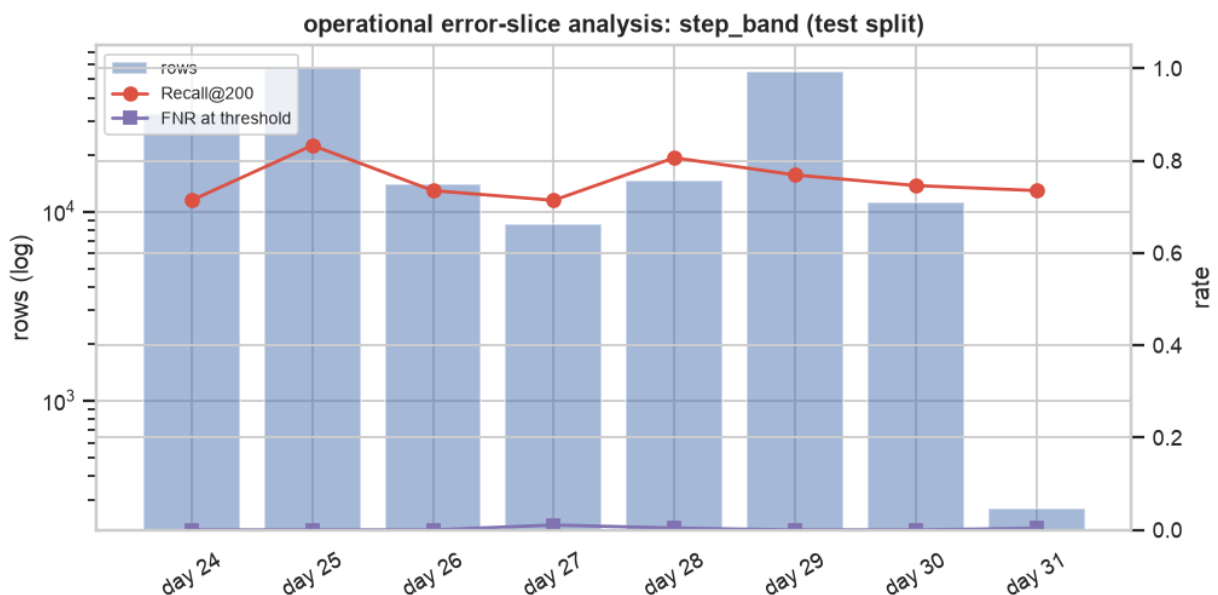
Operational slice by type; slices without positives show no recall point. This is an error-slice view, not a protected-group fairness measure.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.



Operational slice by amount_band; slices without positives show no recall point. This is an error-slice view, not a protected-group fairness measure.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.



Operational slice by orig_balance_band; slices without positives show no recall point. This is an error-slice view, not a protected-group fairness measure.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.



Operational slice by step_band; slices without positives show no recall point. This is an error-slice view, not a protected-group fairness measure.
 Synthetic PaySim data; educational decision-support prototype; no AML determination.

Observations (task T078)

Observations from the tables above (test split, K = 200, raw-score threshold 0.9719). These describe where the ranking's capacity constraint lands; they are not fairness measurements across protected groups, because the dataset has none.

- Type.** Positives exist only in CASH_OUT (1,060) and TRANSFER (1,060). Recall@200 is 0.77 and 0.74 respectively with Precision@200 = 1.0 in both; FNR at threshold is 0.28% and 0.19%. CASH_IN, DEBIT and PAYMENT contain no positives and produce zero false positives at threshold.
- Amount band (edges fitted on training).** The queue's capacity is consumed by the largest transactions: Recall@200 is 0.96 in the high band and 0.87 in very high, 0.54 in mid, and 0.00 in low (106 positives) and very low (74 positives). Threshold FNR in those two bands is 0.0% and 4.1%, so the model does flag most of them; they lose the ranking contest for the 200 daily slots to larger transactions with slightly higher scores and wait unreviewed.

- **Origin balance band.** The same pattern: Recall@200 rises from 0.00 (Q1, 72 positives) and 0.05 (Q2) to 0.82 (Q3) and 0.87 (Q4). The zero band has 10 positives with FNR 0.30: the three zero-amount, zero-balance CASH_OUT rows below threshold.
- **Simulated day.** Recall@200 varies only with the positive count (0.71–0.83); FNR is at most 1.1% (day 27). Day 31 is a partial day of 272 rows, all positives.

The operational consequence is that a fixed daily capacity ranked by this model systematically defers small-value and low-balance positives. That is a property of ranking under capacity, not of any person or group, and the mitigations below address it as such.

Limitations

- **No demographic fairness measurement is possible.** PaySim carries no age, gender, ethnicity, nationality or socioeconomic attributes and no proxy columns (availability record above). Nothing in this analysis speaks to disparate treatment of people; the slices are transaction properties.
- **Class imbalance handling.** Training prevalence is 0.077%; the model uses class weighting only, no resampling. Validation (0.83%) and test (1.09%) prevalence are more than ten times higher because the simulator injects positives at a constant rate (DQ-10), so probabilities calibrated on validation are not calibrated for the training regime.
- **Leakage controls.** Every fitted transform and estimator was fitted on training rows only (fit-scope records); aggregates are causal; the test split was scored once after the operating point was frozen. Post-transaction fields are used deliberately under the batch-triage framing and are labelled batch-only.
- **Overfitting evidence.** Validation PR-AUC 1.0000 and test PR-AUC 1.0000 [1.0000, 1.0000]; the validation-to-test gap is nil for the released model. This reflects separability of the generator rather than generalisation to new behaviour.
- **Synthetic-label validity and simulator artifacts.** The label is produced by simulator rules. The model's two dominant features (balance-arithmetic reconciliation and account emptied to zero) are bookkeeping artifacts of those rules (DQ-03, DQ-05, DQ-06); permutation importance shows nothing else matters to it.
- **Transferability.** Perfect precision at capacity and near-total dependence on two artifact features would not survive contact with real transaction data. Results cannot establish actual fraud or AML detection effectiveness, fairness, or regulatory suitability.

Mitigations

Concrete, feasible for this prototype's context; each names the mechanism, the owner role and the trigger that would invoke it.

- **Capacity-aware slot reservation** (mechanism: reserve a configurable share of the daily K, e.g. 20%, for the highest-scored transactions in under-reviewed amount and balance bands; owner: financial-crime operations lead; trigger: any band's Recall@K falls below half of the overall Recall@K for two consecutive weeks, as it does here for low, very low, Q1 and Q2).
- **Threshold adjustment by review of misses** (mechanism: lower the medium-priority threshold so the five kinds of edge case seen here are labelled medium rather than low; owner: model owner; trigger: FNR at threshold

above 1% in any slice, as on day 27 and in the `zero` balance band).

- **Feature governance for artifact reliance** (mechanism: retrain on the `strict_pretx` set, which reached PR-AUC 0.9995 on test without post-transaction fields, and monitor the permutation importance of any single feature; owner: model owner; trigger: one feature's permutation drop exceeds 0.3, as both bookkeeping flags do today).
- **Reweighting or resampling** (mechanism: per-band sample weights during training so small-value positives are not ranked below large-value ones by default; owner: model owner; trigger: the slot reservation above proves insufficient after one review cycle).
- **Post-processing audit of the queue** (mechanism: weekly report of Recall@K and FNR by slice with the tables above regenerated; owner: model risk reviewer; trigger: scheduled).
- **Monitoring plan** (mechanism: score distribution drift, prevalence per period, Recall@K on labelled batches, and slice tables; owner: model owner; trigger: scheduled weekly, and on any change of data source).

Governance-Controlled Fairness Audit Plan

Required before any use on real transaction data; none of it can be performed on PaySim.

Element	Requirement
Data	Customer records with lawfully obtainable, consented sensitive attributes (or validated proxies) joined to transactions under data-protection review; a documented sampling frame for reviewer decisions and outcomes.
Metrics	Demographic parity difference, equalized odds difference, disparate impact ratio (functions implemented in <code>aml_triage.fairness.demographic</code>) at the operating point and at K; per-group Recall@K, FNR and FPR; calibration by group.
Owners	Model owner (produces metrics), model risk reviewer (independent challenge), compliance/financial-crime lead (accepts or rejects), data-protection officer (attribute handling).
Cadence	Before deployment; then quarterly and after any retraining, threshold change or data-source change; ad hoc on complaint or regulator request.
Human-in-the-loop	Investigators retain the decision and override right; the audit reviews override rates by group as an additional signal.
Outputs	Signed audit record, mitigation decisions with triggers, and a re-run of this report's slice tables on the real data.

7. Limitations

Consolidated from the data-quality review (DQ-01..DQ-13), the model comparison, the explainability review and the Bias & Fairness Analysis. Each item names the evidence.

Data

- **Synthetic origin.** PaySim is generated by an agent-based simulator seeded with aggregate statistics; it is not real SME, corporate or Philippine banking data (Section 2).
- **Constant fraud injection (DQ-10).** Positives arrive at 216–320 per simulated day while legitimate volume swings from about 575,000 to 272 transactions per day, so prevalence is 0.077% in training but 0.83% in validation and 1.09% in test (Section 2, `eda_04`, `eda_05`).
- **Label-correlated artifacts (DQ-03, DQ-05, DQ-06).** All 16 zero-amount rows are positives; simulated positives have exact balance bookkeeping while most normal rows do not; merchant destinations carry no balance state. Features built from these are kept under the batch-triage framing but dominate the released model (Section 5).
- **Two positive types only (DQ-09).** No positives exist in CASH_IN, DEBIT or PAYMENT.
- **No demographics (DQ-13).** No sensitive attributes or proxies exist, so demographic fairness cannot be measured (Section 6).

Modelling

- **Class imbalance** is handled with class weighting only (no resampling); probabilities calibrated on validation are not calibrated for the training regime (Section 4).
- **Leakage controls** are in place (train-only fitting recorded for every transform and estimator; causal aggregates; single-touch test with an audited access record), but post-transaction fields are used deliberately and are labelled batch-only.
- **Overfitting evidence is nil on this data** (validation PR-AUC 1.0000, test 1.0000); this reflects separability of the generator, not generalisation to new behaviour.
- **Artifact dependence.** Permutation importance shows the released model relies almost entirely on two bookkeeping features (Section 5). A model trained without post-transaction fields reaches PR-AUC 0.9995 on test, so the behavioural signal exists, but the released model takes the shortcut.
- **Capacity bound.** Recall@K equals K divided by the positives per period for every strong model; the metric cannot separate them here, and it systematically defers small-value positives (Section 6).
- **Small evaluation windows.** Six validation and eight test review periods; one test period (day 31) is a partial day of 272 rows, all positives. Bootstrap CIs resample rows and understate uncertainty about future periods.

Transferability and governance

- Results cannot establish actual fraud or AML detection effectiveness, fairness, or regulatory suitability. Perfect precision at capacity would not survive contact with real transaction data.
- Any real use requires the governance-controlled validation and fairness audit described in Section 6, a human-in-the-loop workflow with override, and monitoring of drift, prevalence, Recall@K and slice error rates.

8. Reproducibility

Environment

Python 3.11.12 (`.python-version`), dependencies pinned to exact versions in `requirements.txt` (compiled from `requirements.in` with `uv`), development tooling in `requirements-dev.txt`. Global seed 42 is set in `configs/base.yaml` and propagated to every splitter, sampler, estimator, search, and SHAP sample. OMP threads 4, `n_jobs` 4.

Commands (from a clean clone)

```
make setup                # venv, pinned deps, editable install, pre-commit hooks
make data                  # fetch PaySim (Kaggle API token or manual), verify SHA-256
python -m aml_triage validate-schema  # V2: schema confirmed on 6,362,620 rows
python -m aml_triage profile          # reports/data_quality.*
python -m aml_triage data-dictionary
make pipeline EVALUATE_FLAGS='--force-reevaluate --reason "clean-clone reproducibility run"'
                                # split -> features -> selection/PCA -> train -> compare -> tune
                                # -> operating point -> freeze -> evaluate (audited) -> select
                                # -> reproduce-check -> explain -> fairness -> build-report
make test                  # test suite incl. leakage, guard, vocabulary checks
make report && make slides
```

The test split may be scored once per configuration; a second `evaluate --split test` is refused unless `--force-reevaluate --reason "..."` is given, and the reason is appended to `data/processed/test_access.json`. The split manifest and that record are tracked in git, so a clean clone starts frozen and already evaluated: `split` and `freeze` verify that the recomputed partition and operating point are identical to the sealed ones (and refuse otherwise), and the `evaluate` flag above records the rerun in the audit trail.

Clean-clone check (validation task SC-003, run 2026-09-06)

The commands above were executed in a fresh clone of the repository on the same machine (Python 3.11.12, pinned requirements, PaySim re-downloaded and checksum-verified, license re-confirmed as CC BY-SA 4.0 from the Kaggle API metadata). Outcome:

- Every one of the 301 metric values in the released bundle's `metrics.json` (validation and test, point estimates and bootstrap intervals, operating-point counts) reproduced exactly; only the four provenance fields (fit seconds and timestamps) differed.
- The clean-clone estimator and the released estimator produce identical scores on all 181,068 validation rows (maximum absolute difference 0.0) and identical training curves.
- The temporal split, the selected feature set, the tuned hyperparameters, and the sealed operating point (threshold 0.971931, $K = 200$, cutoff 0.999977) were all regenerated identically. The Balanced Random Forest search's best cross-validation score differed only in the ninth decimal place (parallel summation order); its chosen parameters were identical.
- `pipeline.joblib` is not byte-identical across runs because the bundle embeds its version string and the operating point's `chosen_at` timestamp; `pipeline.sha256` therefore identifies one build, while equality of scores is the reproducibility criterion. No `data`, `.env`, or `.joblib` files are tracked; `detect-secrets` found nothing.

The check also surfaced a defect that this run fixed: the CI smoke pipeline had shared three tracked paths with the real run and had overwritten the operating point file, the feature registry's `selected` markers, and the README

tolerance line with smoke values. The released bundle's snapshots and the audit record still held the real values and no report or deck had used the smoke values; the files were restored and every isolated configuration now refuses to write them.

Measured tolerance (validation task V13)

`python -m aml_triage reproduce-check` refit the released candidate twice in fresh processes: **exact** — maximum absolute difference 0.00e+00 in per-row scores and 0.00e+00 against the released bundle's validation metrics (`reports/reproducibility.json`).

Artifacts and versions

Artifact	Location
Released bundle	<code>models/20260904T225142-0dc8f82-hgb/</code> (<code>pipeline.sha256</code> , <code>config_snapshot.yaml</code> , <code>metrics.json</code> , <code>feature_list.json</code> , <code>model_card.md</code>); <code>models/LATEST</code>
Split manifest	<code>data/processed/split_manifest.json</code> (config hash <code>sha256:d099f2d0939d...</code>)
Test-access record	<code>data/processed/test_access.json</code> (state <code>evaluated</code> , first evaluated 2026-09-04T22:51:34+00:00, 0 re-evaluations, 1 audited re-freeze)
Fit-scope records	<code>data/processed/feature_pipeline_*.fitscope.json</code> (all <code>fitted_on: ["train"]</code>)
Operating point	<code>configs/operating_point.yaml</code> (chosen on validation, frozen 2026-09-04T22:22:38+00:00)
Tuned parameters	<code>configs/models/*.tuned.yaml</code> ; search logs under <code>models/tuning/</code>
Data provenance	<code>data/README.md</code> , <code>configs/data_source.yaml</code> (SHA-256 verified on every fetch)

Raw data, processed parquet files and `.joblib` binaries are never committed; `make pipeline` regenerates them, and `reproduce-check` plus the clean-clone check above show that a regenerated model scores identically to the released one.

Educational decision-support prototype trained on synthetic PaySim data. Outputs are risk scores and review priorities that help human investigators decide what to review first. This system makes no fraud or AML determination and performs no automatic blocking, account closure, customer risk rating, or regulatory reporting. Results on synthetic data do not establish real-world detection effectiveness, fairness, or regulatory suitability.